

Topic 9

Chemical Reactions and Energy

Chemistry in Context

Self-heating can Camping is one of the most popular outdoor activities among the youth, but some people do not enjoy it because of the bulky cooking equipment they have to carry with them. To solve this problem, self-heating cans are developed. A self-heating can of drink or food can be re-heated without external heat source or power. A self-heating can consists of two containers. A container for the drink or food surrounds a container of heating agent (e.g. calcium oxide) which is separated from a compartment of water by a thin breakable membrane. When the user pushes on the bottom of the can, a rod pierces the membrane, allowing the water and heating agent to mix. The resulting reaction releases heat and thus warms the drink or food surrounding it.

Do you recall?

1 What fuel do we usually use during camping?
Liquefied butane

2 Write the chemical equation for the reaction between calcium oxide and water.
$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(s)$

3 Suggest another type of chemical reactions that releases heat.
Combustion reaction / neutralization reaction

You are going to study energy changes in chemical reactions.

Unit

34 Energy changes in chemical reactions

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Nature of energy
- Enthalpy changes of reactions
- Standard enthalpy changes of reaction, formation, combustion and neutralization
- Calorimetry

34.1 What is energy?

Heat is a form of energy. We can define energy as 'the capacity to provide heat or to do work'.

When we move an object in the direction of the applied force, for example pushing a load, we have done work.

Energy = heat + work

Energy may take a number of different forms (Fig. 34.1), including:

- sound energy;
- potential energy;
- nuclear energy;
- chemical energy;
- heat;
- electrical energy;
- light energy;
- kinetic energy.

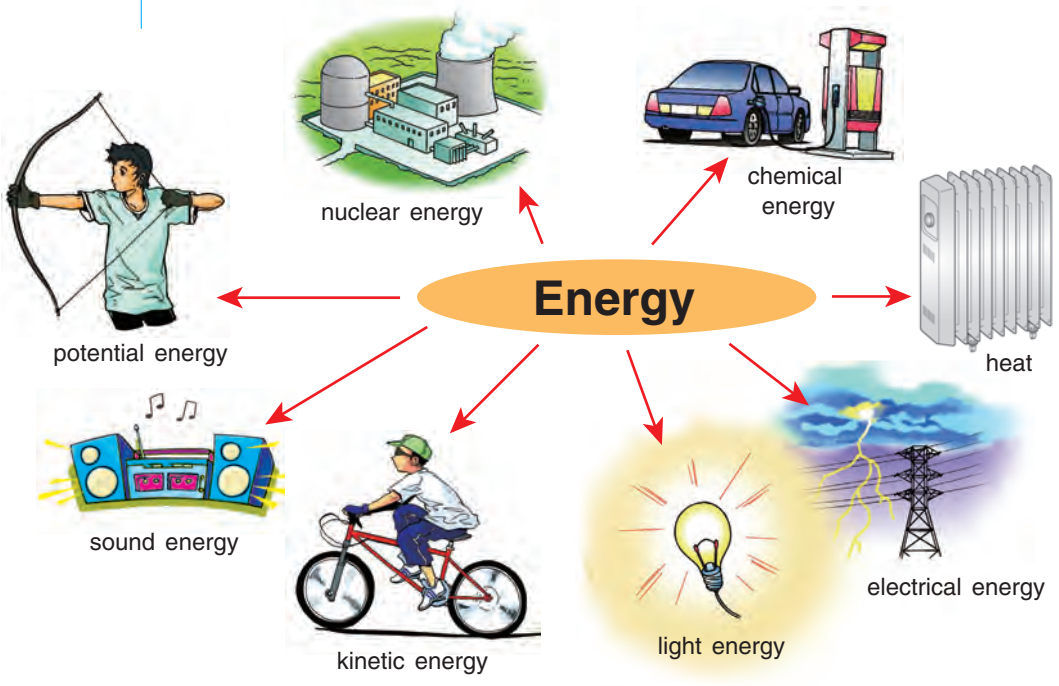


Fig. 34.1 Different forms of energy

Unit of energy

The unit of energy is **joule** (symbol: J). We may consider one joule as the quantity of heat that would raise the temperature of 1 g of water by 0.239 °C.



Fig. 34.2 Have you ever dashed across hot beach sand so that you could stand comfortably in the cool water?

A 1 °C interval on the Celsius scale is equivalent to an interval of 1 Kelvin (symbol: K) on the absolute temperature scale.

34.2 Specific heat capacity

Have you ever dashed across hot beach sand so that you could stand comfortably in the cool water? Both the water and the sand have been heated by the sun for the same amount of time, but why is the temperature of the sand so much higher? This difference in temperature is caused by a difference in the specific heat capacity of water and sand.

✓ The **specific heat capacity** (symbol: c) of a substance is the amount of heat required to raise the temperature of 1 g of the substance by 1 K (or 1 °C).

Notice that specific heat capacity has the units J g^{-1} K^{-1}.

The specific heat capacity of water is 4.18 J g^{-1} K^{-1}. That means it takes 4.18 J of heat to raise the temperature of 1 g of water by 1 K.

Table 34.1 lists the specific heat capacities of some common substances.

Table 34.1

Specific heat capacities of some common substances

Substance	Specific heat capacity (J g^{-1} K^{-1})
Water	4.18
Ethanol	2.41
Aluminium	0.90
Copper	0.39
Glass	0.84
Silicon dioxide (major component of sand)	0.84

joule 焦耳　　specific heat capacity 比熱容



Fig. 34.3 Large lakes can moderate the air temperature of the surrounding areas because of the high specific heat capacity of water

As indicated in Table 34.1, the specific heat capacity of water is considerably higher than that of most other substances. One of the most significant properties of water is that it absorbs a lot of heat to make itself get hot. The high specific heat capacity of water has a great deal to do with regulating temperature in the environment. Oceans and lakes can moderate the air temperature in surrounding areas (Fig. 34.3). The human body, which is about 60% water, is able to maintain a steady internal temperature under changing outside conditions.

We can calculate the amount of heat required to raise the temperature of a given substance as follows:

✓ Amount of heat required
= mass x specific heat capacity x rise in temperature
= m x c x ΔT
where ΔT is the temperature change.

The Greek letter Δ (delta) is used in science and mathematics to signify 'change in'.

Example 34.1

Q Suppose 200 cm^3 of boiling water are needed to make a cup of tea. Calculate the amount of heat required for making a cup of tea if the initial temperature of the water is 17.5 °C.

(Specific heat capacity of water = 4.18 J g^{-1} K^{-1}; density of water = 1.00 g cm^{-3})

A Volume of water = 200 cm^3

Mass of water = 200 cm^3 x 1.00 g cm^{-3}
= 200 g

Amount of heat required = m x c x ΔT
= 200 g x 4.18 J g^{-1} K^{-1} x 82.5 K
= 69.0 kJ

∴ 69.0 kJ of heat are required to make the cup of tea.

Initial temperature of water = 17.5 °C
Final temperature of water = 100.0 °C
ΔT = (100.0 – 17.5) °C
= 82.5 °C (or 82.5 K)

Heat capacity

✓ The **heat capacity** of a substance is the amount of heat required to raise the temperature of the substance by 1 K (or 1 °C).

We can express the relationship between the heat capacity, heat required and the rise in temperature of a substance by the following equation:

$$\text{Heat capacity} = \frac{\text{heat required}}{\text{rise in temperature}}$$

Notice that heat capacity has the units J K^{-1}.

The greater the mass of a substance, the greater is its heat capacity. The amount of heat required to produce a given temperature change of the substance is also greater. For example, a bathtub full of water has a greater mass and heat capacity than a coffee cup full of water (Fig. 34.4). Thus, it takes far more heat to raise the temperature of a bathtub of water by 1 °C than that required to raise the temperature of a coffee cup of water by 1 °C.

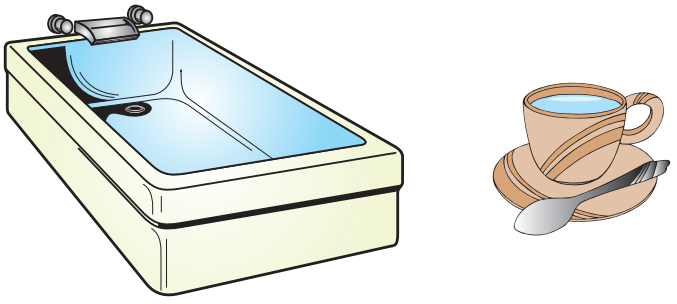


Fig. 34.4 The bathtub full of water has a greater heat capacity than a coffee cup full of water

Practice 34.1

Refer to page T10.

1 In an experiment, a burning fuel raised the temperature of 50.0 cm^3 of water from 16.0 °C to 56.0 °C. How much heat was taken in by the water?
8.36 kJ
(Specific heat capacity and density of water are 4.18 J g^{-1} K^{-1} and 1.00 g cm^{-3} respectively.)

2 Aqueous ethane-1,2-diol is commonly used in car radiators as an antifreeze and coolant. A 50.0% ethane-1,2-diol solution in a radiator has a specific heat capacity of 3.50 J g^{-1} K^{-1}. If 4.00 kg of the solution take in 252 kJ of heat, what was the temperature change of the solution?
18.0 K

heat capacity 熱容量



Fig. 34.5 The chemical energy of the propane fuel in the barbecue is converted to heat and light

34.3 The system and the surroundings

One important characteristic of energy is that it is always conserved. We can neither create nor destroy energy. Energy can only be converted from one form into another.

✓ The **law of conservation of energy** states that energy cannot be created or destroyed; it can only be converted from one form to another.

Think about the combustion of propane gas in a barbecue (Fig. 34.5). Chemical energy stored in propane is converted into heat and light. However, the total amount of energy remains constant before and after the combustion reaction.

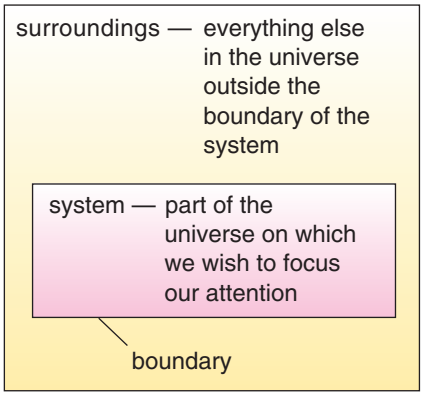


Fig. 34.6 The definitions of the system and the surroundings

To help analyze energy conversions involving chemical reactions, chemists find it useful to divide the universe into two parts: the **system** and its **surroundings**.

The system is the part of the universe on which we wish to focus our attention. The system is separated from its surroundings by a boundary. Everything else in the universe outside the boundary of the system is called the surroundings (Fig. 34.6). Take the example of burning propane gas in a barbecue, the system is composed of the reactants (propane gas and oxygen gas) and the products (carbon dioxide gas and water vapour). The system releases heat and it is transferred to the food (the surroundings).

34.4 Internal energy of a system

The **internal energy** of a system has two components: kinetic energy and potential energy. The kinetic energy is due to the motion of particles (atoms or molecules or ions) and the movement of electrons within particles. The potential energy is determined by

- the attractive interactions between electrons and nuclei;
- the repulsive interactions between electrons and between nuclei in individual particles;
- interactions between particles.

law of conservation of energy 能量守恆定律　system 體系　surroundings 外界　internal energy 內能

It is impossible to measure all these contributions accurately, so we cannot calculate the internal energy of a system with any certainty. However, we can measure changes in internal energy of a system by experiment.

Any change in the internal energy of a system is equal to the difference between its initial and final values.

$$\Delta E = E_f - E_i$$

where ΔE is the change in internal energy of the system,
E_f is the final value of its internal energy,
E_i is the initial value of its internal energy

During a reaction, the change in internal energy of a system is equal to the sum of the heat released or taken in by the system (symbol: q) and the work done by or on the system (symbol: w). That is,

$$\Delta E = q + w$$

For example, consider the reaction between magnesium and dilute hydrochloric acid carried out in the set-up shown in Fig. 34.7. Heat is released and work is done by the hydrogen gas (product) in expanding against the atmospheric pressure.

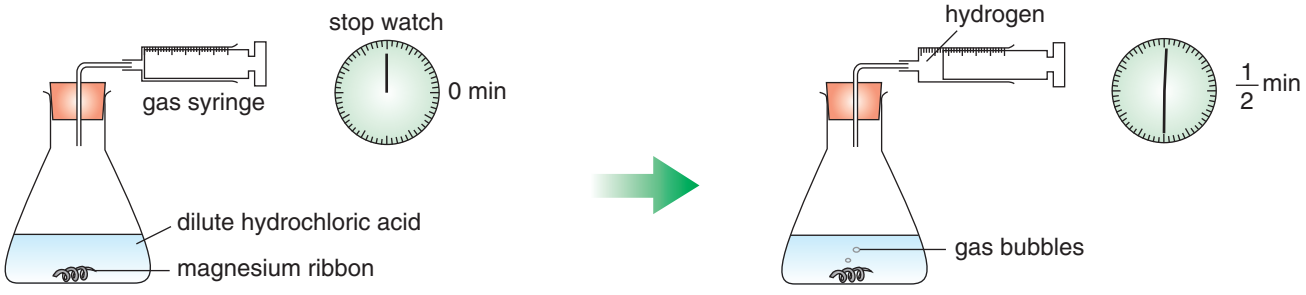


Fig. 34.7 Reaction between magnesium and dilute hydrochloric acid

Thus, the change in internal energy of the system is equal to the sum of heat released and work done by the system.

As we normally carry out reactions in the laboratory at constant pressure, the heat released or taken in by a system (q) kept at constant pressure is given a special name, **enthalpy change** (symbol: ΔH).

✓ An enthalpy change, ΔH, is the heat released or taken in during any change in a system, provided that the system is kept at constant pressure.

enthalpy change 焓變

The enthalpy of a substance, sometimes called its heat content, is an indication of its energy content. Consider the following reaction:

$$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$$

Magnesium has a certain enthalpy, and so does oxygen. We can represent the total enthalpy of the reactants (magnesium and oxygen) by the symbol H_r. The product, magnesium oxide, also has its enthalpy, represented by the symbol H_p. The enthalpy change of the reaction is given by the equation:

$$\Delta H = H_p - H_r$$

The difference between the enthalpy change (ΔH) and internal energy change (ΔE) is usually small in most cases. Thus, the value of ΔH can give a very good indication of the internal energy change of a system during a reaction.

34.5 Enthalpy change of an exothermic reaction

34.1
Classifying whether reactions are exothermic or endothermic.

✓ In an **exothermic reaction**, heat is released to the surroundings.

Video
Finding pairs of chemicals that may be used to make heat packs and cold packs

When quicklime (CaO) is added to water, heat is released and the temperature of the water rises (Fig. 34.8). The reaction between quicklime and water is an exothermic reaction.

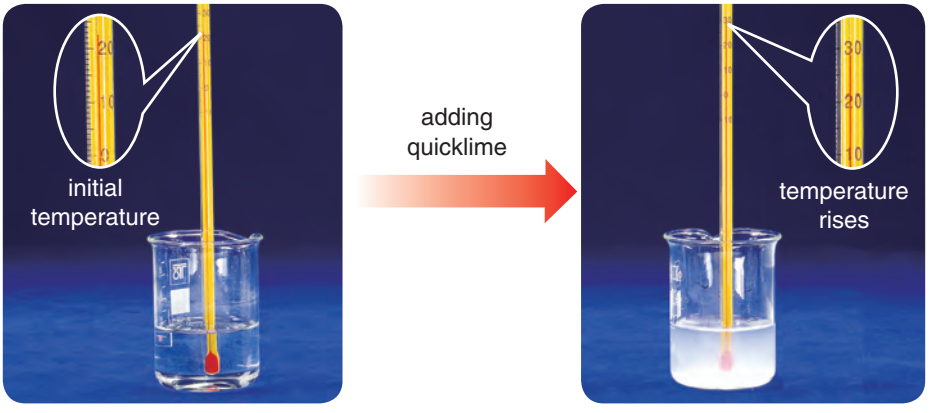


Fig. 34.8 When quicklime is added to water, heat is released and the temperature rises

exothermic reaction 放熱反應

N2 Refer to page T10.

Do you know

Heat packs N2

One common example of an exothermic reaction is the reaction that occurs in the heat packs used by athletes to minimize muscle sprains.

CE 2009 Paper 1 Q3(a) & (c)

The heat in one type of such packs comes from the oxidation of iron, which is an exothermic reaction. A damp mixture of iron filings with salt and charcoal is contained in a perforated bag. The mixture is kept in a sealed bag which keeps the oxygen out. The pack is activated by either squeezing or breaking the seal, allowing the oxidation reaction to take place. The reaction is speeded up by the presence of the salt and charcoal.

$$4Fe(s) + 3O_2(g) \longrightarrow 2Fe_2O_3(s)$$



Fig. 34.10 A heat pack

The total enthalpy of the products is less than that of the reactants. Thus, the products are written at a lower level of the enthalpy level diagram.

enthalpy level diagram 焓級圖

Other examples of exothermic reactions include: N1

- the combustion of fuels (Fig. 34.9);
- the oxidation of carbohydrates in the bodies of animals and plants (respiration); and
- neutralization reactions.



Fig. 34.9 Combustion of fuel produces enough energy to launch a rocket

An enthalpy level diagram for an exothermic reaction

In an exothermic reaction, the total enthalpy of the products (H_p) is less than that of the reactants (H_r). Hence the enthalpy change of the reaction ΔH (= $H_p - H_r$) is a negative quantity, $\Delta H < 0$.

We can represent the enthalpy change of an exothermic reaction by an **enthalpy level diagram** (Fig. 34.11).

DSE 2013 Paper 1B Q7(a)(ii)

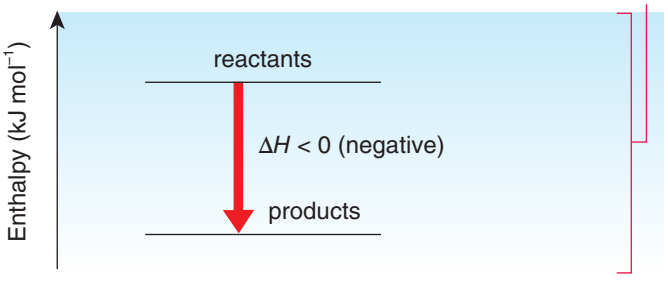


Fig. 34.11 An enthalpy level diagram for an exothermic reaction

N1 The following processes are exothermic:
- freezing of water;
- condensation of steam;
- dissolving sodium hydroxide pellets in water;
- fermentation of glucose;
- the reaction of lime with dilute hydrochloric acid;
- $CaCO_3(s) + 2H^+(aq) \longrightarrow Ca^{2+}(aq) + H_2O(l) + CO_2(g)$

34.6 Enthalpy change of an endothermic reaction

✓ In an **endothermic reaction**, heat is taken in from the surroundings.

When sodium carbonate is added to ethanoic acid, heat is taken in and the temperature falls (Fig. 34.12). The reaction between sodium carbonate and ethanoic acid is an endothermic reaction.

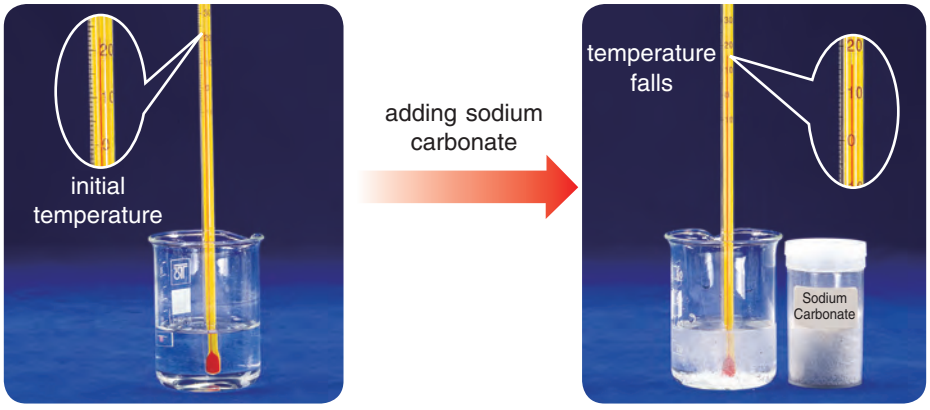


Fig. 34.12 The reaction between ethanoic acid and sodium carbonate produces a temperature fall

N3 The following processes are endothermic:
- melting of ice;
- evaporation of ethanol;
- sublimation of iodine;
- electrolysis of water;
- cracking of naphtha.

Other examples of endothermic reactions include: N3

- the thermal decomposition of limestone (Fig. 34.13);
- photosynthesis (where the energy is supplied by sunlight); and
- heating blue hydrated copper(II) sulphate crystals to obtain white anhydrous copper(II) sulphate.

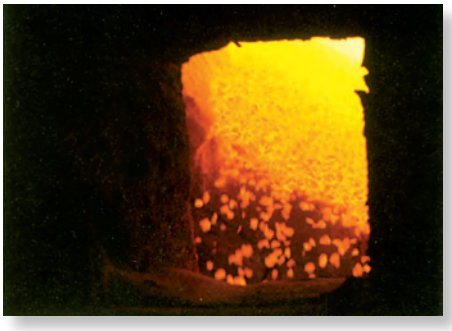


Fig. 34.13 Heating limestone in a lime kiln to produce quicklime (calcium oxide)

endothermic reaction 吸熱反應

An enthalpy level diagram for an endothermic reaction

In an endothermic reaction, the total enthalpy of the products (H_p) is greater than that of the reactants (H_r). Hence the enthalpy change of the reaction ΔH (= $H_p - H_r$) is a positive quantity, $\Delta H > 0$.

We can represent the enthalpy change of an endothermic reaction by an enthalpy level diagram (Fig. 34.14).

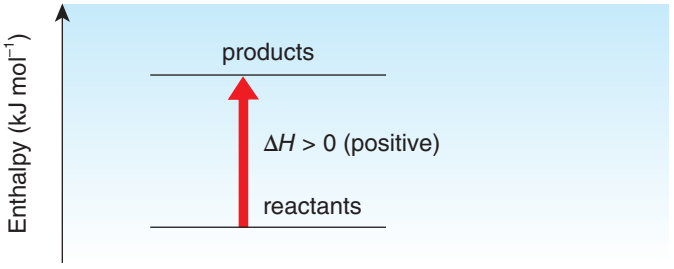


Fig. 34.14 An enthalpy level diagram for an endothermic reaction

The total enthalpy of the products is greater than that of the reactants. Thus, the products are written at an upper level of the enthalpy level diagram.

Practice 34.2

1 Identify each of the following processes as exothermic or endothermic.

a) Burning of magnesium in air
Exothermic

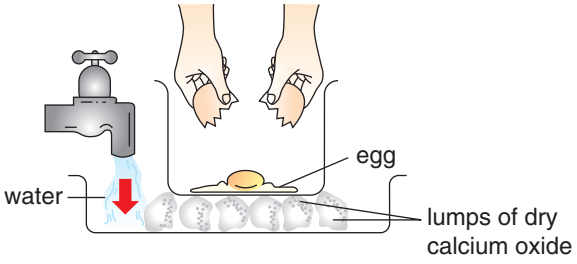
b) Cracking of distilled fractions of petroleum
Endothermic

c) Evaporation of sea water
Endothermic

d) Reaction between sodium and water
Exothermic

2 Use the diagram to explain how the egg can be cooked.

The reaction between dry calcium oxide and water is exothermic. Heat is released when water is added to the dry calcium oxide. Thus, the egg is cooked by the heat released.



34.7 Enthalpy changes during physical and chemical changes

So far, we have discussed enthalpy changes that occur during chemical changes. Actually almost every change in a system involves an enthalpy change. The change can be either physical or chemical.

N4 Refer to page T11.

Enthalpy change during a physical change — melting of ice N4



Fig. 34.15 Ice melting is an endothermic process

At 0 °C and atmospheric pressure, ice melts to form water (Fig. 34.15). The melting process is a physical change. Measurements show that for every mole of ice that melts, 6.01 kJ of heat are taken in by the ice. Because the process occurs at constant pressure, the heat change (q) is equal to the enthalpy change (ΔH).

The equation for this physical change is:

$$H_2O(s) \longrightarrow H_2O(l) \qquad \Delta H = +6.01 \text{ kJ mol}^{-1}$$

Ice melting is an endothermic process. Hence ΔH is a positive quantity.

The 'per mole' in the units for ΔH means that this is the enthalpy change when 1 mole of ice is converted to 1 mole of water.

We can represent the enthalpy change involved in the process by the enthalpy level diagram shown in Fig. 34.16.

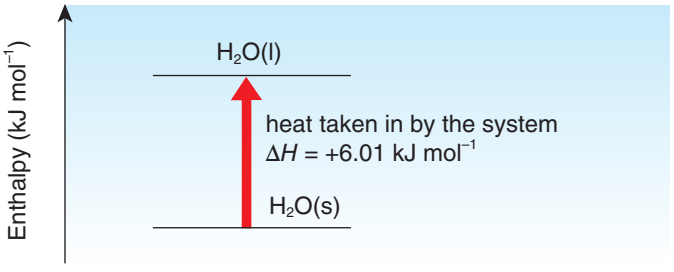


Fig. 34.16 An enthalpy level diagram showing the enthalpy change involved in the melting of ice to form water

Do you know

Cold packs

Cold packs are often used by athletes to relieve pain. Ammonium nitrate is often used in cold packs because it absorbs a lot of heat when it dissolves in water. In a cold pack, ammonium nitrate and water are kept in separate chambers. When the pack is needed, the chambers are broken and the ammonium nitrate dissolves in the water, absorbing heat and making the pack as cold as 0 °C. Dissolving ammonium nitrate in water is an endothermic process. The dissolving process is a physical change, NOT a chemical change.

Enthalpy change during a chemical change — combustion of methane

Consider the combustion of methane (CH_4), which is the principal component of natural gas. We know from experience that burning natural gas releases heat to the surroundings.

The equation for this chemical change is:

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H = -890 \text{ kJ mol}^{-1}$$

The combustion of methane is an exothermic process. Hence ΔH is a negative quantity. The equation tells us that when 1 mole of methane reacts with 2 moles of oxygen to yield 1 mole of carbon dioxide and 2 moles of liquid water, 890 kJ of heat are released to the surroundings.

We can represent the enthalpy change involved in the process by the enthalpy level diagram shown in Fig. 34.17.

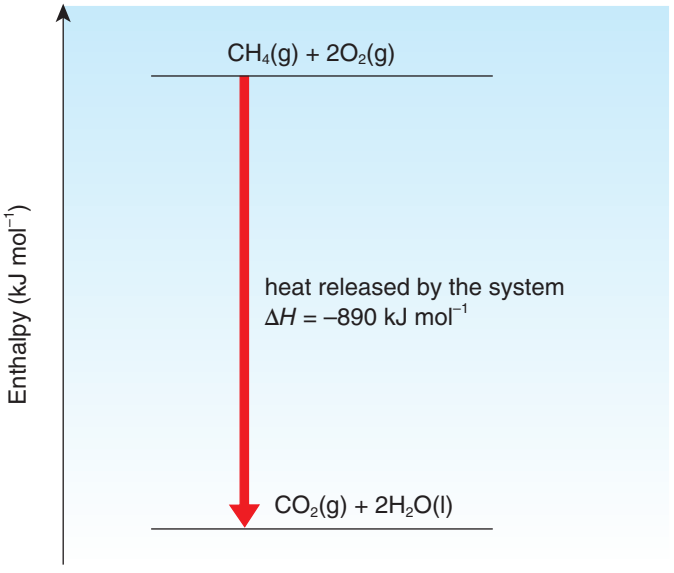
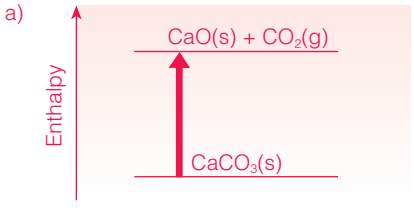


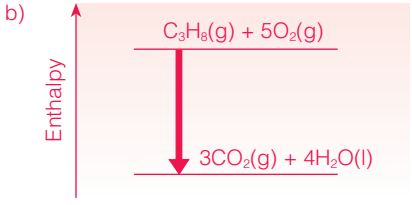
Fig. 34.17 An enthalpy level diagram showing the enthalpy change involved in the combustion of methane

Practice 34.3

a)



b)



Practice 34.3

Draw enthalpy level diagrams for

a) the thermal decomposition of limestone;

b) the combustion of propane gas.

34.8 Thermochemical equations

The equation for the combustion of methane shown above is called a **thermochemical equation**. It shows the enthalpy change as well as the mole relationship.

It is essential to specify a balanced equation when quoting the enthalpy change of a reaction. The following guidelines are helpful in writing and interpreting thermochemical equations.

1 **Specifying the physical states of the reactants and products in a thermochemical equation**

When writing thermochemical equations, we must always specify the physical states of all reactants and products. For example, hydrogen burns in oxygen to form water but there are two possibilities:

- forming liquid water

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l) \qquad \Delta H = -286 \text{ kJ mol}^{-1}$$

- forming steam

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g) \qquad \Delta H = -242 \text{ kJ mol}^{-1}$$

The difference in ΔH represents the amount of heat required to turn one mole of liquid water into steam.

2 **Multiplying ΔH by the same factor when multiplying both sides of an equation**

The enthalpy change for a reaction depends on the quantities of materials reacting. For example, the combustion of 2 moles of methane in 4 moles of oxygen releases 1 780 kJ of heat.

$$2CH_4(g) + 4O_2(g) \longrightarrow 2CO_2(g) + 4H_2O(l)$$
$$\Delta H = 2(-890 \text{ kJ})$$
$$= -1\,780 \text{ kJ}$$

Thus, when we multiply both sides of an equation by a factor, ΔH must also be multiplied by the same factor.

During calculations, if we obtain the enthalpy change of a reaction by multiplying the value of ΔH (in kJ mol^{-1}) with the number of moles of substances involved, the unit of the enthalpy change obtained should be 'kJ'.

thermochemical equation 熱化學方程式

3 Changing the sign of ΔH when reversing an equation

When we reverse an equation, we change the roles of the reactants and products. Consequently, the magnitude of ΔH for the equation remains the same, but its sign changes.

For example, the synthesis of water from its elements is an exothermic reaction.

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l) \qquad \Delta H = -286 \text{ kJ mol}^{-1}$$

Now consider the reverse of this synthesis. Electricity can be used to decompose water into its elements. For this reaction to occur, water must take in 286 kJ mol^{-1} of energy. The reaction is endothermic.

$$H_2O(l) \longrightarrow H_2(g) + \frac{1}{2}O_2(g) \qquad \Delta H = +286 \text{ kJ mol}^{-1}$$

Example 34.2

Q Given the following thermochemical equation:

$$\frac{1}{2}N_2(g) + O_2(g) \longrightarrow NO_2(g) \qquad \Delta H = +34 \text{ kJ mol}^{-1}$$

a) Determine the enthalpy change (ΔH) of the reaction:

$$2NO_2(g) \longrightarrow N_2(g) + 2O_2(g)$$

b) Calculate the amount of heat involved when 2.76 g of $NO_2(g)$ decompose to give $N_2(g)$ and $O_2(g)$.

(Relative atomic masses: N = 14.0, O = 16.0)

A a) Multiplying both sides of the given equation by 2 gives the following equation:

$$N_2(g) + 2O_2(g) \longrightarrow 2NO_2(g) \qquad \Delta H = 2(+34 \text{ kJ}) = +68 \text{ kJ}$$

When the reaction proceeds in the reverse direction, the magnitude of ΔH for the equation remains the same, but its sign changes.

i.e. $2NO_2(g) \longrightarrow N_2(g) + 2O_2(g) \qquad \Delta H = -68 \text{ kJ}$

b) When 2 moles of $NO_2(g)$ decompose to give $N_2(g)$ and $O_2(g)$, 68 kJ of heat are released.

i.e. when 1 mole of $NO_2(g)$ decomposes to give $N_2(g)$ and $O_2(g)$, 34 kJ of heat are released.

Number of moles of $NO_2(g)$ decomposed $= \dfrac{2.76 \text{ g}}{(14.0 + 2 \times 16.0) \text{ g mol}^{-1}} = 0.0600 \text{ mol}$

Heat released when 2.76 g of $NO_2(g)$ decomposed = 0.0600(34 kJ) = 2.04 kJ

Practice 34.4

Refer to page T12.

1 For the decomposition of hydrogen peroxide, it is known that:

$$H_2O_2(l) \longrightarrow H_2O(l) + \frac{1}{2}O_2(g) \qquad \Delta H = -98.2 \text{ kJ mol}^{-1}$$

a) Draw an enthalpy level diagram for the decomposition of hydrogen peroxide.

b) Using the given information, determine the enthalpy change (ΔH) for the reaction:

$$2H_2O(l) + O_2(g) \longrightarrow 2H_2O_2(l)$$

+196 kJ

2 The combustion of ethanol occurs according to the following equation:

$$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(g) \qquad \Delta H = -1\,368 \text{ kJ mol}^{-1}$$

Calculate the amount of heat released when 3.00 g of ethanol burn completely.

89.2 kJ

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

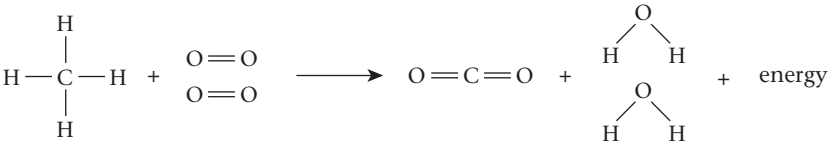
Animation
Energy changes associated with breakage and formation of chemical bonds

34.9 Explaining energy changes — breakage and formation of chemical bonds

DSE 2015 Paper 1B Q8(b)(i) & (ii)

Consider the combustion of methane gas. It can be summarized in the following way:

$$\underset{\text{methane gas}}{CH_4(g)} + \underset{\text{oxygen gas}}{2O_2(g)} \longrightarrow \underset{\text{carbon dioxide gas}}{CO_2(g)} + \underset{\text{water}}{2H_2O(l)} + \text{energy}$$



The reaction releases considerable heat. To gain a better understanding of the energy change involved, imagine that the reaction takes place in two distinct steps. In the first step, suppose that all the chemical bonds in one CH_4 molecule and two O_2 molecules are broken. Such a bond-breaking step is an energy-requiring process.

Step 1:

$$\text{energy} + CH_4 + 2O_2 \longrightarrow C + 4H + 4O$$

To complete the reaction, suppose that the separate atoms now rearrange themselves to form the product molecules. Such a bond-forming step is an energy-releasing process.

Step 2:

$$C + 4H + 4O \longrightarrow CO_2 + 2H_2O + \text{energy}$$

The amount of energy released in the bond-forming step is greater than the amount of energy used in the bond-breaking step. So, the overall reaction is exothermic (Fig. 34.18).

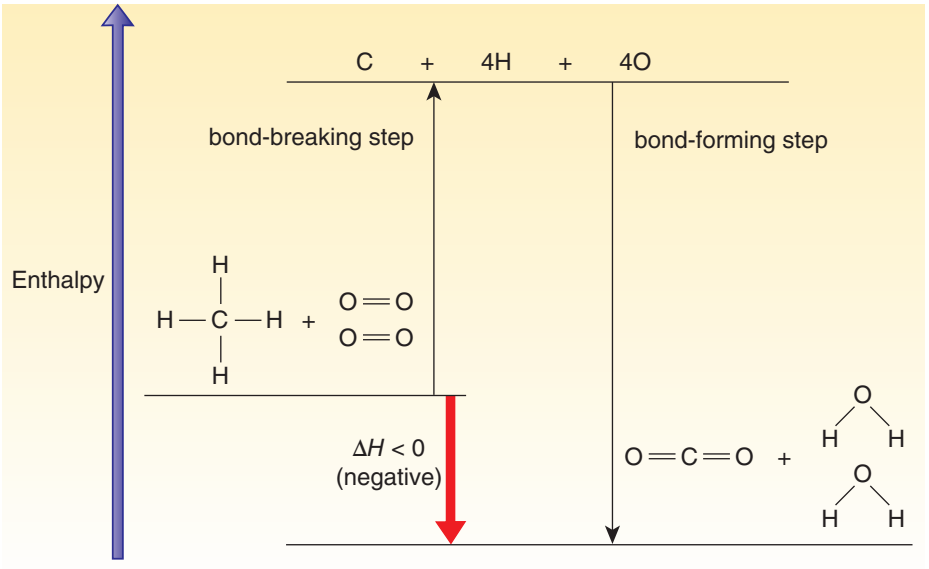


Fig. 34.18 An enthalpy level diagram showing the enthalpy change involved in the combustion of methane

In an exothermic reaction, the energy required to break the existing bonds is less than the energy released from forming new bonds.

In an endothermic reaction, the energy required to break the existing bonds is greater than the energy released from forming new bonds.

Practice 34.5

Inside car engines, the air gets so hot that nitrogen and oxygen react together, forming nitrogen monoxide. This reaction is endothermic.

$$N_2(g) + O_2(g) \longrightarrow 2NO(g)$$

Explain, in terms of bond breaking and forming, why the reaction is endothermic.

The amount of energy released in the bond-forming step is less than the amount of energy used in the bond-breaking step.

34.10 Standard conditions for measuring enthalpy changes

Notice that the value of an enthalpy change is affected by temperature, pressure and concentration of solutions.

To make any comparison of enthalpy changes a fair comparison, chemists have defined a set of **standard conditions** under which enthalpy changes are measured. The standard conditions are:

- at a temperature of 25 °C (298 K);
- at a pressure of 1 atmosphere (atm);
- all the substances involved are in their standard states; i.e. the most stable physical state at 25 °C and 1 atm;
- all the solutions involved have a concentration of 1 mol dm^{-3}.

For example, carbon dioxide is a gas while water is a liquid at 25 °C and 1 atm.

Any enthalpy change measured under these conditions is described as a standard enthalpy change and given the symbol $\Delta H^{\ominus}$.

In case of elements which exist as different **allotropes**, and compounds which exist as different **polymorphs**, the most stable form at 25 °C and 1 atm is chosen as the standard state.

Different forms of the same compound are called polymorphs.

The most stable state of a substance is when all its internal energy has been released. Hence substances are more stable if they are at a lower level of the enthalpy level diagram.

AL 2006 Paper 1 Q2(a)(iii)

Consider the two allotropes of carbon, graphite and diamond. At 25 °C and 1 atm, graphite is the more stable form of carbon than diamond. Consequently, $\Delta H^{\ominus}$ values for reactions involving carbon should relate to the allotrope graphite, rather than diamond.

Sulphur also exists in many allotropic forms, but the most stable form at 25 °C and 1 atm is rhombic sulphur.

Practice 34.6

Consider the following reaction:

$$C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H^{\ominus} = -1\,560 \text{ kJ mol}^{-1}$$

a) What does the symbol 'Δ' mean?
Change in

b) What does the symbol '*H*' mean?
Enthalpy

c) What does the symbol '⊖' mean?
c) The enthalpy change is measured under standard conditions
i.e. at a temperature of 25 °C;
at a pressure of 1 atm.

d) What does the minus sign indicate?
Heat is released from the system to the surroundings.

standard conditions 標準條件　allotrope 同素異型體　polymorph 多晶型物

34.11 Standard enthalpy change of reaction

✓ The **standard enthalpy change of reaction**, $\Delta H_r^\ominus$, is the enthalpy change when molar quantities of reactants as stated in the specified equation react together under standard conditions.

The amount of heat released or taken in during a chemical reaction depends on the quantity of reactants. To avoid any confusion about quantities, we need to give an equation.

For example, the equation

$$C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H_r^\ominus = -1\,560 \text{ kJ}$$

shows us the enthalpy change when 1 mole of ethane reacts with $\frac{7}{2}$ moles of oxygen to give carbon dioxide and water. In this case 1 560 kJ of heat are released.

However, if we write the equation as

$$2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(l) \qquad \Delta H_r^\ominus = -3\,120 \text{ kJ}$$

Here 2 moles of ethane react with 7 moles of oxygen to give carbon dioxide and water. In this case 3 120 kJ of heat are released.

AL 2008 Paper 2 Q1(c)(i)

34.12 Standard enthalpy change of formation

✓ The **standard enthalpy change of formation** of a substance, $\Delta H_f^\ominus$, is the enthalpy change when one mole of the substance is formed from its elements in their standard states.

For example, the standard enthalpy change of formation of water is -286 kJ mol^{-1}. When we make one mole of water from hydrogen and oxygen, 286 kJ of heat are released. This is summed up as:

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l) \qquad \Delta H_f^\ominus[H_2O(l)] = -286 \text{ kJ mol}^{-1}$$

The units of enthalpy change of formation are kJ mol^{-1}. Notice that 'per mole' refer to the formation of one mole of the compound, not to the quantity of elements reacting.

Notice that when we write the thermochemical equation for the enthalpy change of formation of a substance, the coefficient of that substance must always be 1. Other substances in the equation may have fractional coefficients as a result.

standard enthalpy change of reaction 標準反應焓變　　standard enthalpy change of formation 標準生成焓變

Fig. 34.19 shows an enthalpy level diagram for the standard enthalpy change of formation of water.

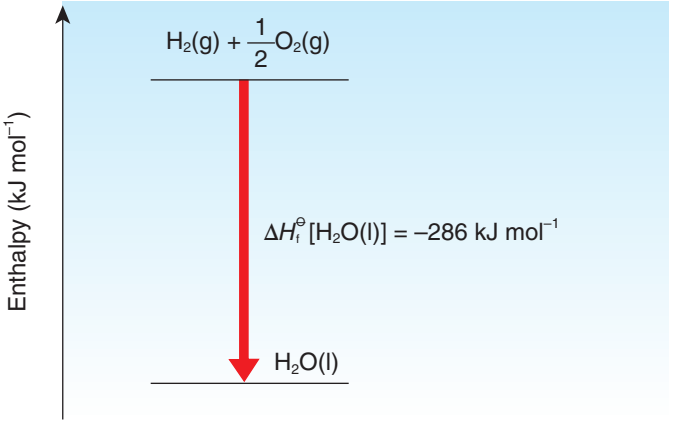


Fig. 34.19 An enthalpy level diagram for the standard enthalpy change of formation of water

Notice that the 'reaction' that forms a substance from its constituent elements can be (and often is) hypothetical. Practically, carbon and hydrogen react with oxygen to give carbon dioxide and water respectively.

The standard enthalpy change of formation of ethanol, $\Delta H_f^{\ominus}[C_2H_5OH(l)]$, is –278 kJ mol^{-1}.

$$2C(\text{graphite}) + 3H_2(g) + \frac{1}{2}O_2(g) \longrightarrow C_2H_5OH(l)$$

$$\Delta H_f^{\ominus}[C_2H_5OH(l)] = -278 \text{ kJ mol}^{-1}$$

By convention, the standard enthalpy change of formation of any element in its standard state is zero. Take the element carbon as an example. At 25 °C and 1 atm, graphite is a more stable allotropic form of carbon than diamond, so we have

$$\Delta H_f^{\ominus}[C(\text{graphite})] = 0 \quad \text{but} \quad \Delta H_f^{\ominus}[C(\text{diamond})] \neq 0$$

Table 34.2 lists the standard enthalpy changes of formation ($\Delta H_f^{\ominus}$) of some common substances.

Table 34.2

Standard enthalpy changes of formation $\Delta H_f^{\ominus}$ of some common substances

Process of formation	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$	–242
$Mg(s) + C(\text{graphite}) + \frac{3}{2}O_2(g) \longrightarrow MgCO_3(s)$	–1 113
$\frac{1}{2}N_2(g) + O_2(g) \longrightarrow NO_2(g)$	+39.3

Continued on next page

Process of formation	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$C(graphite) + 2H_2(g) \longrightarrow CH_4(g)$	–74.8
$C(graphite) + O_2(g) \longrightarrow CO_2(g)$	–394
$2C(graphite) + 2H_2(g) \longrightarrow C_2H_4(g)$	+52.2
$C(graphite) + 2H_2(g) + \frac{1}{2}O_2(g) \longrightarrow CH_3OH(l)$	–239

What can enthalpy changes of formation tell us?

Chemists often talk about the stability of a compound in relation to enthalpy changes — but what do they mean? Take hydrogen peroxide as an example. It has a standard enthalpy change of formation of –188 kJ mol^{-1}.

$$H_2(g) + O_2(g) \longrightarrow H_2O_2(l) \qquad \Delta H_f^{\ominus}[H_2O_2(l)] = -188 \text{ kJ mol}^{-1}$$

This tells us that the enthalpy of one mole of hydrogen peroxide is less than the total enthalpy of one mole of hydrogen and one mole of oxygen by 188 kJ. Thus, we can say that hydrogen peroxide is more stable than its constituent elements.

34.13 Standard enthalpy change of combustion

✓ The **standard enthalpy change of combustion** of a substance, $\Delta H_c^{\ominus}$, is the enthalpy change when one mole of the substance is completely burnt in oxygen under standard conditions.

For example, the standard enthalpy change of combustion of octane, one of the alkanes found in petrol, is –5 470 kJ mol^{-1}.

$$C_8H_{18}(l) + \frac{25}{2}O_2(g) \longrightarrow 8CO_2(g) + 9H_2O(l)$$

$$\Delta H_c^{\ominus}[C_8H_{18}(l)] = -5\,470 \text{ kJ mol}^{-1}$$

DSE 2014 Paper 1B Q6(a)(iii)

standard enthalpy change of combustion 標準燃燒焓變

The term 'completely burnt' in the definition is important, since many elements form more than one oxide. For example, carbon may burn to form two types of oxide:

$$C(\text{graphite}) + \frac{1}{2}O_2(g) \longrightarrow CO(g) \qquad \Delta H^{\ominus} = -111 \text{ kJ mol}^{-1}$$

$$C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H^{\ominus} = -394 \text{ kJ mol}^{-1}$$

Complete combustion of carbon produces carbon dioxide, so only the enthalpy change of the second process is regarded as the standard enthalpy change of combustion of carbon.



Fig. 34.20 Combustion reactions are always exothermic

The standard enthalpy changes of combustion ($\Delta H_c^{\ominus}$) of some common substances are listed in Table 34.3. As combustion reactions are always exothermic (Fig. 34.20), all values of the standard enthalpy changes of combustion are negative.

Table 34.3

Standard enthalpy changes of combustion $\Delta H_c^{\ominus}$ of some common substances

Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})	Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})
$H_2(g)$	–286	$C_2H_6(g)$	–1 560
C(graphite)	–394	$C_8H_{18}(l)$	–5 470
CO(g)	–283	$CH_3OH(l)$	–726
$CH_4(g)$	–890	$C_2H_5OH(l)$	–1 368
$C_2H_4(g)$	–1 411	$C_6H_{12}O_6(s)$	–2 803

The standard enthalpy change of combustion of carbon (graphite) corresponds to the standard enthalpy change of the following reaction:

$$C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H^{\ominus} = -394 \text{ kJ mol}^{-1}$$

The standard enthalpy change of the above reaction may also be called the standard enthalpy change of formation of carbon dioxide. Thus, different enthalpy changes may refer to the same thermochemical equation (with the same ΔH value).

Consider another example,

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l) \qquad \Delta H^{\ominus} = -286 \text{ kJ mol}^{-1}$$

The standard enthalpy change of the above reaction may be called the standard enthalpy change of combustion of hydrogen or the standard enthalpy change of formation of water.

Practice 34.7

1 Represent each of the following by a thermochemical equation:

a) the standard enthalpy change of formation of aluminium oxide is –1 676 kJ mol^{-1};

b) the standard enthalpy change of combustion of propan-1-ol is –2 021 kJ mol^{-1}.

2 Classify the enthalpy change of each of the following reactions as $\Delta H_r^\ominus$, $\Delta H_f^\ominus$ or $\Delta H_c^\ominus$:

a) $MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$

b) $S(s) + O_2(g) \longrightarrow SO_2(g)$

c) $C_6H_{12}O_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$

Practice 34.7

1 a) $2Al(s) + \frac{3}{2}O_2(g) \longrightarrow Al_2O_3(s)$ $\Delta H_f^\ominus = -1\ 676\ kJ\ mol^{-1}$

b) $C_3H_7OH(l) + \frac{9}{2}O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$ $\Delta H_c^\ominus = -2\ 021\ kJ\ mol^{-1}$

2 a) $\Delta H_r^\ominus$

b) $\Delta H_f^\ominus[SO_2(g)]$ or $\Delta H_c^\ominus[S(s)]$

c) $\Delta H_c^\ominus[C_6H_{12}O_6(s)]$

34.14 Standard enthalpy change of neutralization

Reference Website

Refer to the following website for the animations on heat changes when mixing different acids and alkalis:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/animations/Rxnheat2.htm
(accessed July 2014)

✓ The **standard enthalpy change of neutralization**, $\Delta H_n^\ominus$, is the enthalpy change when an acid reacts with an alkali to form one mole of water under standard conditions.

Acid-alkali neutralizations in aqueous solutions are always exothermic. Consider the reaction between dilute hydrochloric acid and sodium hydroxide solution:

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l) \qquad \Delta H_n^\ominus = -57.1\ kJ\ mol^{-1}$$

In this reaction, all the reactants are present as ions, so:

$$H^+(aq) + Cl^-(aq) + Na^+(aq) + OH^-(aq) \longrightarrow Na^+(aq) + Cl^-(aq) + H_2O(l)$$

Notice that the ions reacting are $H^+(aq)$ ions and $OH^-(aq)$ ions, $Na^+(aq)$ ions and $Cl^-(aq)$ ions do not take part in the reaction. So, the neutralization reaction can be simply written as:

$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$$

standard enthalpy change of neutralization 標準中和焓變

Table 34.4 shows the standard enthalpy changes of neutralization ($\Delta H_n^\ominus$) of some acid-alkali reactions. Notice that the first four reactions in the table have approximately the same $\Delta H_n^\ominus$ value. A strong monobasic acid and a strong alkali are involved in each case. Thus, the same chemical change

$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$$

is taking place in each case.

The standard enthalpy change of neutralization involving a strong monobasic acid and a strong alkali is about –57.1 kJ mol^{-1}.

Table 34.4

Standard enthalpy changes of neutralization $\Delta H_n^\ominus$ of some common acids and alkalis

Acid	Alkali	$\Delta H_n^\ominus$ (kJ mol^{-1})
HCl	NaOH	–57.1
HCl	KOH	–57.2
HNO_3	NaOH	–57.2
HNO_3	KOH	–57.1
HCl	NH_3	–52.2
H_2SO_4	NaOH	–64.8
H_2SO_4	KOH	–64.8
CH_3COOH	NaOH	–55.2
CH_3COOH	NH_3	–51.5

For neutralization in which either the acid or alkali or both are weak, the standard enthalpy change of neutralization will be less than –57.1 kJ mol^{-1}.

For example,

$$HCl(aq) + NH_3(aq) \longrightarrow NH_4Cl(aq) \qquad \Delta H_n^\ominus = -52.2 \text{ kJ mol}^{-1}$$

This is because some energy is consumed when the weak acid and weak alkali dissociate to give hydrogen ions and hydroxide ions before neutralization.

Do you know

Measuring chemical energy in food

Calorimetry is also used to measure the energy released by burning food. In food science, energy is often measured in calories. One calorie, 1 kcal, is 4.2 kJ.

34.15 Determining enthalpy changes of chemical reactions

Calorimetry is the process of measuring enthalpy changes of chemical reactions or physical changes.

For a reaction that takes place in solution, we can determine the enthalpy change of the reaction by measuring the temperature change of the reaction mixture. A device used for measuring the heat transferred during a reaction is called a **calorimeter**.

A typical calorimeter consists of a vacuum flask or two expanded polystyrene cups with a lid (Figs. 34.21–34.22).

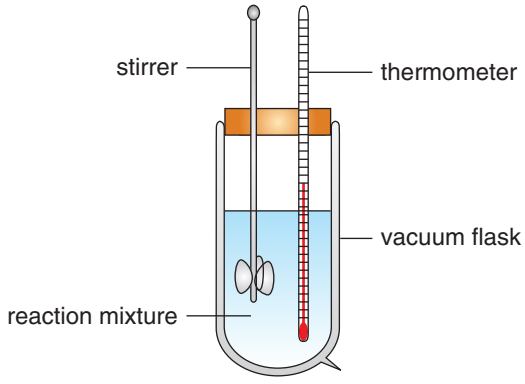


Fig. 34.21 A simple calorimeter consists of a vacuum flask

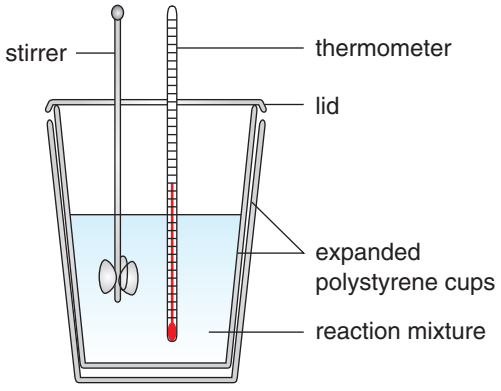


Fig. 34.22 A simple calorimeter consists of two expanded polystyrene cups with a lid

34.2
Determining the enthalpy change of a reaction.

Video
Determining the enthalpy change of a reaction

For the calorimeter consisting of two expanded polystyrene cups, the inner cup is used to hold the reaction mixture and the outer cup is used to provide extra insulation.

Experimental procedures for determining the enthalpy change of neutralization

Suppose we want to determine the enthalpy change of neutralization for the reaction between hydrochloric acid and sodium hydroxide solution:

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

34.3
Determining the enthalpy changes of neutralization.

Video
Determining the enthalpy change of neutralization of the reaction between hydrochloric acid and sodium hydroxide solution

calorimetry 量熱法 calorimeter 熱量計

We can follow the steps listed below:

1 Place 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid in a calorimeter. Measure its temperature.

2 Add 50.0 cm^3 of 1.00 mol dm^{-3} sodium hydroxide solution (at the same temperature) to the acid (Fig. 34.23).

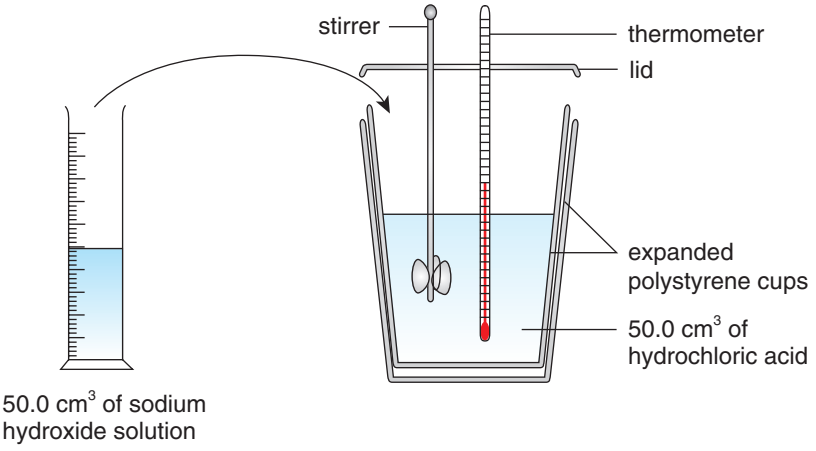


Fig. 34.23 A calorimeter for measuring the standard enthalpy change of neutralization

3 Replace the lid, stir the reaction mixture. Measure the final maximum temperature of the reaction mixture.

As the reaction mixture is a dilute solution, we can take its specific heat capacity and density to be the same as those of water (i.e. specific heat capacity = 4.18 J g^{-1} K^{-1}, density = 1.00 g cm^{-3}). Also, we can assume the heat capacity of the polystyrene calorimeter to be negligible.

Results and calculations

Initial temperature of the acid / alkali = 21.3 °C
Final maximum temperature of the reaction mixture = 27.8 °C
Maximum temperature rise = 6.5 °C

Total volume of the reaction mixture = 50.0 cm^3 + 50.0 cm^3
= 100.0 cm^3

Notice that the total volume of the reaction mixture is 100.0 cm^3. A common mistake is using 50.0 cm^3 in the calculation.

Mass of the reaction mixture = 100.0 g

Amount of heat released during neutralization
= amount of heat gain for the reaction mixture
= $m \times c \times \Delta T$
= 100.0 g x 4.18 J g^{-1} K^{-1} x 6.5 K = 2 720 J = 2.72 kJ

Number of moles of HCl reacted
= Number of moles of NaOH reacted

$$= 1.00 \text{ mol dm}^{-3} \times \frac{50.0}{1\,000} \text{ dm}^3$$

$$= 0.0500 \text{ mol}$$

0.0500 mole of HCl reacted with 0.0500 mole of NaOH to produce 0.0500 mole of H_2O.

When 0.0500 mole of water was formed, 2.72 kJ of heat were released.

Amount of heat released when 1 mole of water is formed

$$= \frac{2.72 \text{ kJ}}{0.0500 \text{ mol}}$$

$$= 54.4 \text{ kJ mol}^{-1}$$

Therefore the enthalpy change of neutralization for the reaction is $-54.4 \text{ kJ mol}^{-1}$.

Neutralization is an exothermic reaction. Hence the value of the enthalpy change of neutralization is negative.

Fig. 34.24 summarizes the steps we use to calculate the enthalpy change of neutralization from experimental results.

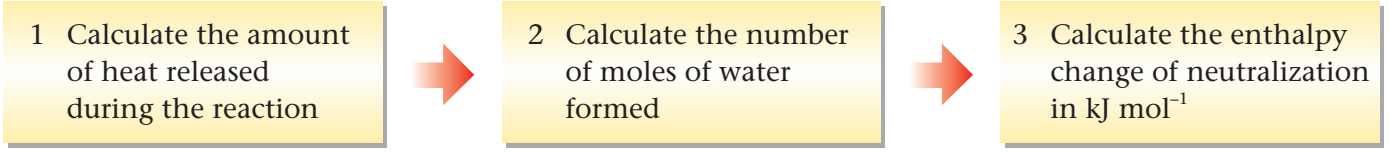


Fig. 34.24 Steps for calculating the enthalpy change of neutralization from experimental results

DSE 2012 Paper 1B Q8(d)(ii)

Sources of inaccuracy in the experiment

The value of the enthalpy change of neutralization ($-54.4 \text{ kJ mol}^{-1}$) determined in the above experiment is less than the theoretical value of $\Delta H_n^{\ominus}$ ($-57.1 \text{ kJ mol}^{-1}$). Sources of inaccuracy in the experiment include:

- heat loss to the surroundings;
- heat capacities of the polystyrene calorimeter, the thermometer and the stirrer being ignored;
- thermometer not being precise enough;
- density and specific heat capacity of the solution not being exactly the same as those of water; and
- the experiment is not carried out under standard conditions.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Suggest ways to make results of the experiment more accurate.
Refer to page T13.

Example 34.3

Q An experiment was carried out to determine the enthalpy change of the following reaction:

$Mg(s) + Cu^{2+}(aq) \longrightarrow Cu(s) + Mg^{2+}(aq)$

50.0 cm^3 of 0.500 mol dm^{-3} $CuSO_4(aq)$ were transferred to an expanded polystyrene cup with negligible heat capacity, and the temperature of the solution was recorded.

0.500 g of magnesium were added to the cup. The mixture was stirred and its highest temperature was recorded. There was a temperature rise of 51.0 °C.

a) Show, by calculation, that Mg is the limiting reactant.

(Relative atomic mass: Mg = 24.3)

b) The mass of the resulting solution is 51.0 g. Assuming that the specific heat capacity of the resulting solution is 4.18 J g^{-1} K^{-1}, calculate the enthalpy change of this reaction, in kJ mol^{-1}.

A a) Number of moles of $CuSO_4$ = 0.500 mol dm^{-3} x $\frac{50.0}{1\,000}$ dm^3

= 2.50×10^{-2} mol

Number of moles of Mg = $\frac{0.500 \text{ g}}{24.3 \text{ g mol}^{-1}}$

= 2.06×10^{-2} mol

According to the equation, 1 mole of Mg reacts with 1 mole of $CuSO_4$. During the reaction, 2.06×10^{-2} mole of Mg reacted with 2.06×10^{-2} mole of $CuSO_4$.

∴ Mg is the limiting reactant.

b) Amount of heat released during the reaction = $m \times c \times \Delta T$

= 51.0 g x 4.18 J g^{-1} K^{-1} x 51.0 K

= 10 900 J

= 10.9 kJ

Enthalpy change of reaction = $\frac{-10.9 \text{ kJ}}{2.06 \times 10^{-2} \text{ mol}}$

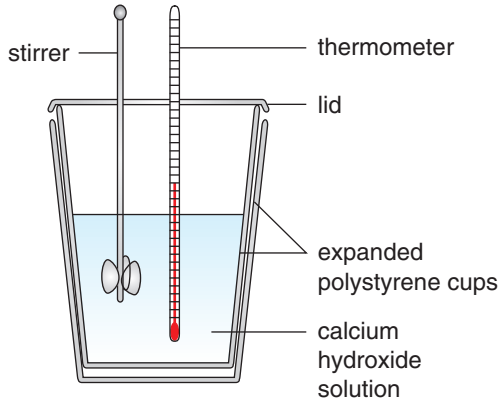
= –529 kJ mol^{-1}

∴ the enthalpy change of the reaction is –529 kJ mol^{-1}.

Example 34.4

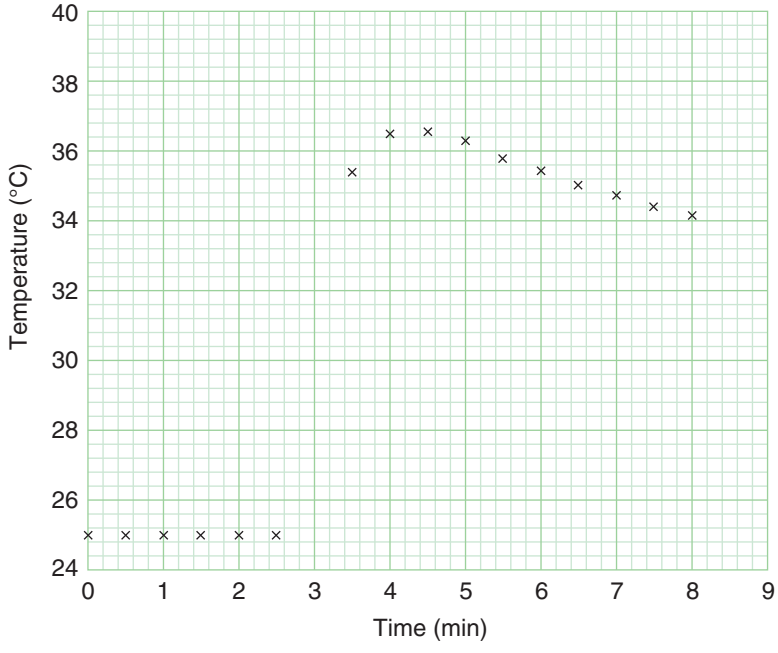
Q Calcium oxide reacts with water to form calcium hydroxide solution.

A student used the following set-up to determine the enthalpy change of the reaction between calcium oxide and water.



The student first placed 100.0 g of water in an expanded polystyrene calorimeter and recorded its temperature at half-minute intervals for 2.5 minutes. At the third minute, the student added 4.00 g of calcium oxide to the water, replaced the lid and stirred. Then he recorded the temperature of the reaction mixture for an additional 5 minutes.

The recordings of temperature are shown in the graph below:



DSE 2013 Paper 1B Q5(a)(i)
ASL 2005 Q4(b)
ASL 2011 Paper 2 Q3(a)(i)
AL 2011 Paper 2 Q3(a)(i)

a) Deduce the greatest temperature change of the reaction mixture.

Continued on next page

b) Calculate the enthalpy change, in kJ mol^{-1}, of the reaction of calcium oxide with water under the experimental conditions.

c) Why did the student measure the temperature of the water for several minutes before adding the calcium oxide?

d) Why was an expanded polystyrene cup, rather than a glass beaker, used as a calorimeter in the experiment?

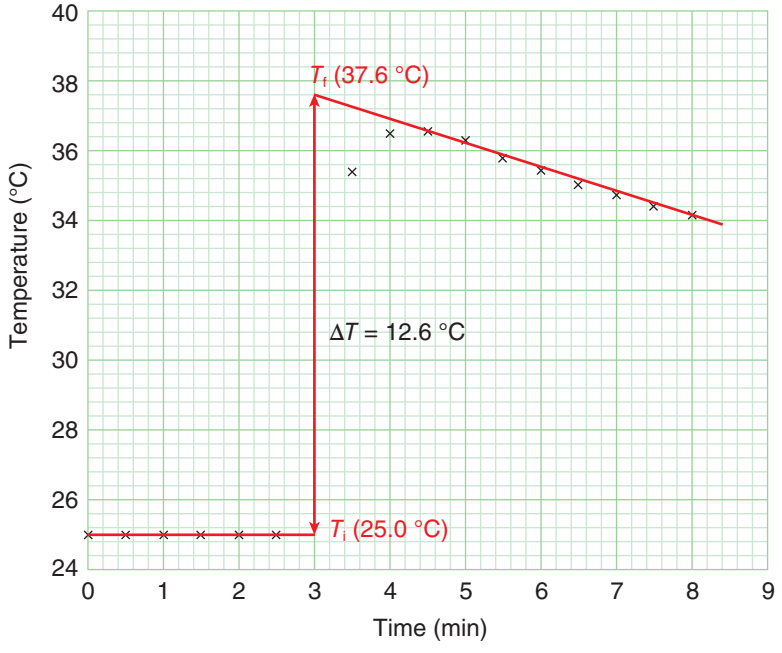
(Relative atomic masses: O = 16.0, Ca = 40.1; you may assume that the specific heat capacity of the calcium hydroxide solution is 4.20 J g^{-1} K^{-1}, and the heat capacity of the expanded polystyrene calorimeter is negligible.)

A1 a) We can determine the maximum temperature rise (ΔT) by the *extrapolation method*:

- Join the points before the addition of CaO(s) using a straight line and extrapolate to the third minute (the time at which CaO(s) is added). Label the temperature at the third minute as T_i (25.0 °C).
- Join the points after the addition of CaO(s) using a straight line and extrapolate back to the third minute. Label the temperature at the third minute as T_f (37.6 °C).
- The separation of the lines at the third minute corresponds to the maximum temperature change for the process (indicated as ΔT on the graph).

Notice that we do not measure the temperature of the reaction mixture right after the addition of CaO(s) at the third minute. This is because the exothermic reaction is vigorous and fast. It is impossible to measure the accurate temperature at that moment. The extrapolation method allows us to find the maximum temperature that would have been reached if there is no heat loss.

Maximum temperature rise $\Delta T = T_f - T_i$
$= 37.6\ °C - 25.0\ °C$
$= 12.6\ °C$



extrapolation method 外推法

Continued on next page

b) Amount of heat released when calcium oxide reacted with water

$= m \times c \times \Delta T$

$= 104.0 \text{ g} \times 4.20 \text{ J g}^{-1} \text{ K}^{-1} \times 12.6 \text{ K}$

$= 5\ 500 \text{ J}$

$$\text{Number of moles of CaO(s) reacted} = \frac{4.00 \text{ g}}{56.1 \text{ g mol}^{-1}}$$

$$= 0.0713 \text{ mol}$$

Enthalpy change of the reaction of calcium oxide with water

$$= \frac{-5\ 500 \text{ J}}{0.0713 \text{ mol}}$$

$= -77.1 \text{ kJ mol}^{-1}$

$\therefore$ the enthalpy change of reaction of calcium oxide with water is $-77.1 \text{ kJ mol}^{-1}$.

c) To determine the initial temperature of the water and to judge whether it was constant.

d) Expanded polystyrene is a good insulator and can minimize heat loss to the surroundings. Glass is not a good insulator compared with expanded polystyrene.

Practice 34.8

Refer to page T13.

1 25.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ potassium hydroxide solution and 25.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ hydrochloric acid, both at the same temperature, were mixed in a calorimeter of negligible heat capacity. A maximum temperature rise of 3.4 °C was recorded.

In a second similar experiment, the hydrochloric acid was replaced by 25.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ ethanoic acid. The maximum temperature rise was 2.6 °C.

a) Calculate the respective enthalpy changes of neutralization of hydrochloric acid and ethanoic acid by potassium hydroxide solution in kJ mol^{-1}.
$-56.9 \text{ kJ mol}^{-1}$; $-43.4 \text{ kJ mol}^{-1}$

b) Explain the difference of the two values calculated in (a).

c) i) What are the possible sources of error in this experiment?

ii) Suggest ways to make the results of the experiment more accurate.

(Specific heat capacity of all reaction mixtures = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$; density of all reaction mixtures = 1.00 g cm^{-3})

Continued on next page

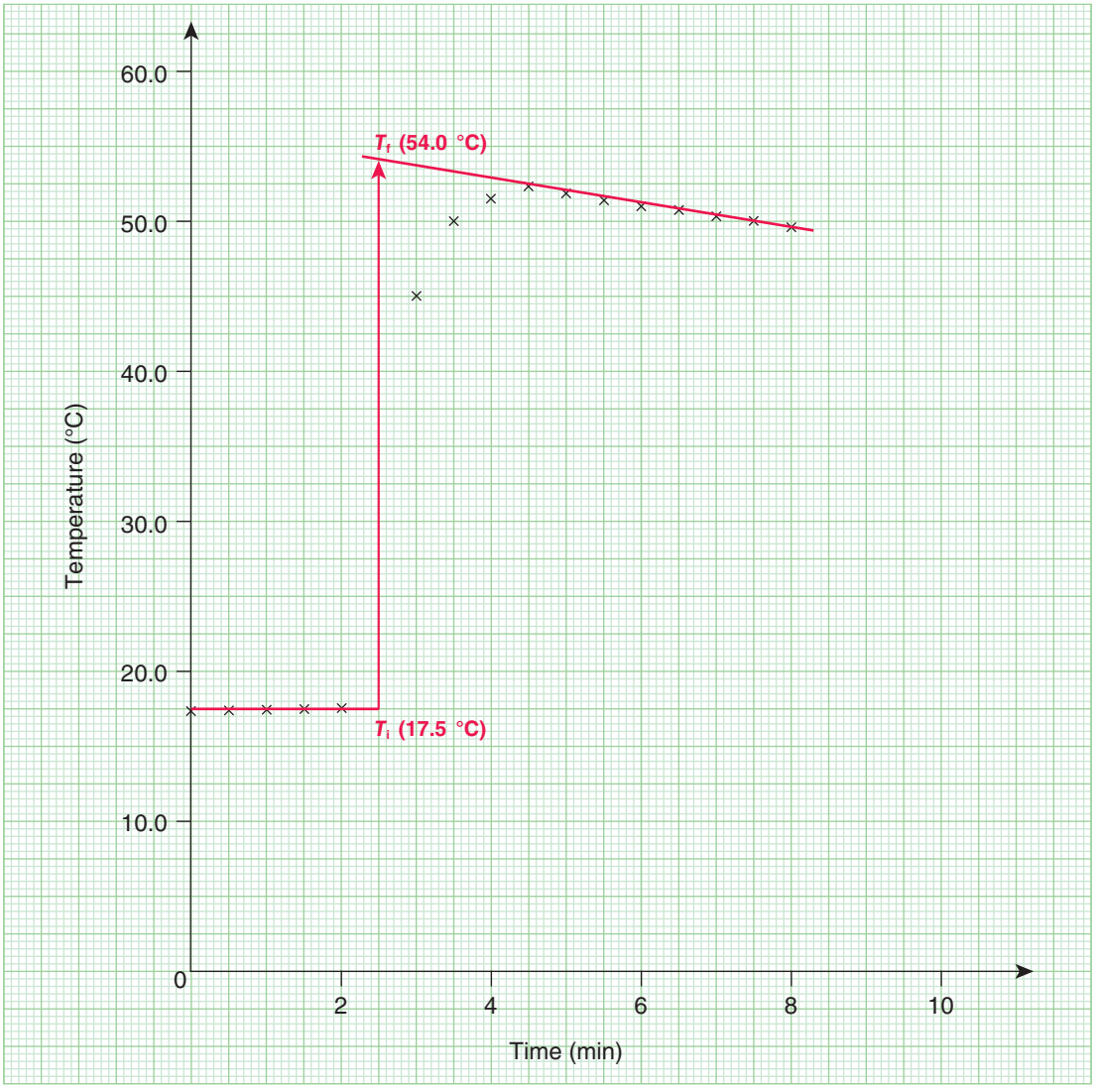
2 An experiment was carried out to determine the enthalpy change of the following reaction:

$Fe(s) + CuSO_4(aq) \longrightarrow Cu(s) + FeSO_4(aq)$

25.0 cm^3 of 1.00 mol dm^{-3} $CuSO_4(aq)$ were transferred to an expanded polystyrene calorimeter with negligible heat capacity. The temperature of the solution was recorded every half minute for 2 minutes.

At 2.5 minutes, excess powdered iron was added to the calorimeter. The mixture was stirred and its temperature was recorded for an additional 5 minutes.

The recordings of temperature are shown in the graph below.



a) Deduce the greatest temperature change of the reaction mixture.
Greatest temperature change of the reaction mixture = (54.0 – 17.5) °C = 36.5 °C

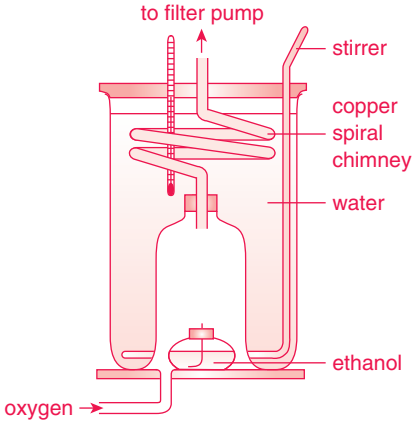
b) Assuming that the specific heat capacity and the density of the mixture are 4.18 J g^{-1} K^{-1} and 1.00 g cm^{-3} respectively, calculate the enthalpy change of this reaction, in kJ mol^{-1}.
–152 kJ mol^{-1}

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

34.4
Determining the enthalpy changes of combustion of some alcohols.

Video
Determining the enthalpy changes of combustion of some alcohols

N5 An improved apparatus: the flame calorimeter
The simple calorimeter shown in Fig. 34.25 is not very efficient and accurate because energy released from combustion of the fuel is lost in heating the apparatus and surroundings. A more effective apparatus, the flame calorimeter, is shown below. The chimney is made of copper, so that heat released from combustion can be transferred effectively to the water. The flame is enclosed so as to reduce heat loss. Oxygen is supplied to ensure complete combustion.



When using a flame calorimeter, we need to know its heat capacity (i.e. the heat required to raise the temperature of the whole apparatus by 1 K). This can be calculated from measurements made using a fuel with known enthalpy change of combustion.

When using the flame calorimeter, the amount of heat gain for it is found as follows:

Amount of heat gain for the calorimeter = heat capacity of the calorimeter x ΔT
where ΔT is the temperature change.

34.16 Determining enthalpy changes of combustion

We can determine the enthalpy change of combustion of a fuel (such as ethanol or oil) by burning a known mass of the fuel in a calorimeter. The amount of heat released in the process is determined by measuring the rise in temperature of the water in the calorimeter.

Fig. 34.25 shows a simple calorimeter used to measure the enthalpy change of combustion of ethanol: N5

$$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$$

We can follow the steps listed below:

1 Record the mass of the spirit lamp containing the ethanol.

2 Record the mass and initial temperature of the water in the calorimeter.

3 Ignite the spirit lamp for about 10 minutes.

4 Record the maximum rise in temperature of the water and the mass of the spirit lamp after the experiment.

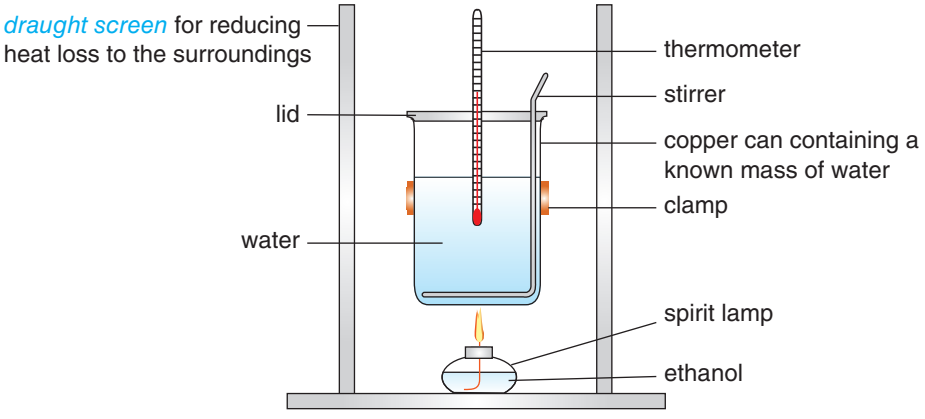


Fig. 34.25 A simple calorimeter used to determine the enthalpy change of combustion of ethanol

draught screen 擋風屏

Results and calculations

Mass of water in the calorimeter = 250.0 g
Mass of spirit lamp before the experiment = 532.68 g
Mass of spirit lamp after the experiment = 531.76 g
Initial temperature of water = 18.3 °C
Final temperature of water = 37.6 °C

Assume that the heat capacity of the calorimeter is negligible.

Given that the specific heat capacity and density of water are 4.18 J g^{-1} K^{-1} and 1.00 g cm^{-3} respectively, calculate the enthalpy change of combustion of ethanol (molar mass = 46.0 g mol^{-1}) as follows:

Temperature change of water = (37.6 – 18.3) °C
= 19.3 °C (or 19.3 K)

Amount of heat released during combustion
= m x c x ΔT
= 250.0 g x 4.18 J g^{-1} K^{-1} x 19.3 K
= 20 200 J
= 20.2 kJ

$$\text{Number of moles of } C_2H_5OH \text{ burnt} = \frac{(532.68 - 531.76)\text{ g}}{46.0\text{ g mol}^{-1}}$$
$$= 0.0200\text{ mol}$$

When 0.0200 mole of ethanol was burnt, 20.2 kJ of heat were released.

Amount of heat released when 1 mole of ethanol is burnt

$$= \frac{20.2\text{ kJ}}{0.0200\text{ mol}}$$
$$= 1\ 010\text{ kJ mol}^{-1}$$

Therefore the enthalpy change of combustion of ethanol is –1 010 kJ mol^{-1}.

Fig. 34.26 summarizes the steps we use to calculate the enthalpy change of combustion from experimental results.

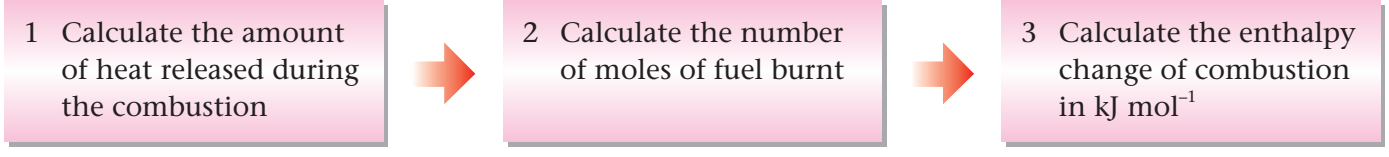


Fig. 34.26 Steps for calculating the enthalpy change of combustion from experimental results

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

In the above example, the experimentally determined value (–1 010 kJ mol^{-1}) is less than the theoretical value of the standard enthalpy change of combustion of ethanol (–1 368 kJ mol^{-1}).

1 Suggest possible sources of error that lead to such a low experimental value.

2 Suggest ways to make the result of the experiment more accurate.

Refer to page T15.

Practice 34.9

The apparatus shown on the right was used to determine the enthalpy change of combustion of propan-1-ol.

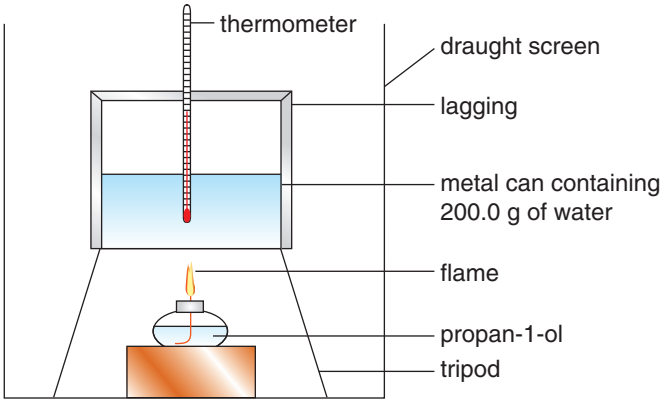
The following results were obtained:

Mass of propan-1-ol burnt = 0.600 g

Mass of water heated = 200.0 g

Initial temperature of water = 21.0 °C

Final temperature of water = 42.9 °C



a) Suggest why the apparatus was lagged and shielded from draughts.
To minimize heat loss to the surroundings.

b) Suggest why the metal can holding the water should be very thin and light.
To minimize the heat absorbed by the metal can.

c) Calculate the enthalpy change of combustion of propan-1-ol. You may assume that the heat capacity of the calorimeter is negligible.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0; specific heat capacity of water = 4.18 J g^{-1} K^{-1})

Practice 34.9

c) Temperature change of water = (42.9 – 21.0) °C
= 21.9 °C

Amount of heat released during combustion of propan-1-ol = 200.0 g x 4.18 J g^{-1} K^{-1} x 21.9 K
= 18 300 J
= 18.3 kJ

Number of moles of propan-1-ol burnt = $\frac{0.600 \text{ g}}{60.0 \text{ g mol}^{-1}}$
= 0.0100 mol

$\Delta H_c = \frac{-18.3 \text{ kJ}}{0.0100 \text{ mol}^{-1}}$
= –1 830 kJ mol^{-1}

∴ the enthalpy change of combustion of propan-1-ol is –1 830 kJ mol^{-1}.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

The energy values of fuels N6 Refer to page T15.

Fossil fuels will eventually run out, so chemists are actively studying alternatives. Possible alternatives include ethanol, methanol and hydrogen.

Theoretical chemists measure the heat released when a fuel burns in kJ mol^{-1} because this compares the heat released by the same number of molecules of each fuel. We can also measure the heat released per gram of fuel burnt, or the energy density of a fuel. This is more appropriate for determining how efficient and effective fuels are.

Some approximate values are given in the table below.



Fig. 34.27 Ethanol can be made from sugar by fermentation

Fuel	Enthalpy change of combustion (kJ mol^{-1})	Mass of 1 mole (g)	Energy density (kJ g^{-1})	Combustion products		Type of flame
				CO_2	H_2O	
Octane (a component of petrol)	–5 470	114	48.0	✔	✔	smoky
Hydrogen	–242	2	121		✔	not smoky
Ethanol	–1 368	46	29.7	✔	✔	not smoky
Methanol	–726	32	22.7	✔	✔	not smoky

Question

1 Hydrogen can be made by the electrolysis of water. Hydrogen can be a renewable fuel if the electricity used to make it is generated from renewable energy resources such as wind or hydroelectric power.

a) Write a chemical equation for the combustion of hydrogen.
$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$

b) How do the product(s) of combustion vary between hydrogen and the other fuels? What environmental significance does this have?
No carbon dioxide is formed.
Carbon dioxide is a greenhouse gas. Emission of carbon dioxide intensifies the greenhouse effect and leads to global warming.

2 Ethanol can be made from sugar by a natural process called fermentation◀. The sugar comes from plants, so ethanol made in this way is a renewable fuel.

Discuss the benefits and drawbacks of using crops to make fuels such as ethanol.
Refer to page T16.

◀ We will discuss fermentation in Topic 10 Rate of reaction and Topic 13 Industrial chemistry.

Have you mastered?

Key terms

joule 焦耳 3 specific heat capacity 比熱容 3 heat capacity 熱容量 5

law of conservation of energy 能量守恆定律 6 system 體系 6 surroundings 外界 6

internal energy 內能 6 enthalpy change 焓變 7 exothermic reaction 放熱反應 8

enthalpy level diagram 焓級圖 9 endothermic reaction 吸熱反應 10

thermochemical equation 熱化學方程式 14 standard conditions 標準條件 18

allotrope 同素異形體 18 polymorph 多晶型物 18

standard enthalpy change of reaction 標準反應焓變 19

standard enthalpy change of formation 標準生成焓變 19

standard enthalpy change of combustion 標準燃燒焓變 21

standard enthalpy change of neutralization 標準中和焓變 23 calorimetry 量熱法 25

calorimeter 熱量計 25

Checklist

After studying this unit, you should be able to

- ☐ explain energy changes in chemical reactions in terms of the concept of conservation of energy;
- ☐ describe enthalpy change (ΔH) as heat change at constant pressure;
- ☐ draw enthalpy level diagrams for exothermic and endothermic reactions;
- ☐ explain the nature of exothermic and endothermic reactions in terms of bond-breaking and bond-forming steps;
- ☐ explain and use standard enthalpy changes of reaction, formation, combustion and neutralization;
- ☐ carry out experimental determination of enthalpy changes using simple calorimetric method;
- ☐ calculate enthalpy changes from experimental results.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 The specific heat capacity (symbol: c) of a substance is the amount of heat required to raise the temperature of 1 g of the substance by 1 K (or 1 °C).

Amount of heat required = mass x specific heat capacity x rise in temperature
$= m \times c \times \Delta T$

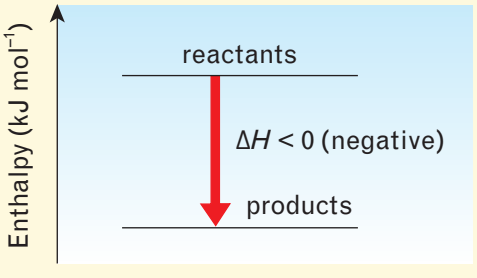
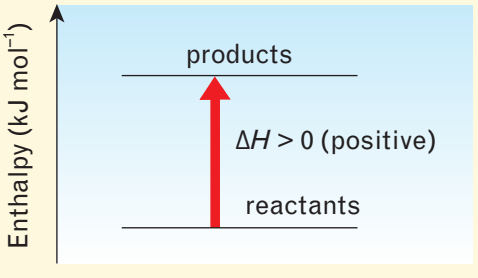
where ΔT is the temperature change.

2 The heat capacity of a substance is the amount of heat required to raise the temperature of the substance by 1 K (or 1 °C)

$$\text{Heat capacity} = \frac{\text{heat required}}{\text{rise in temperature}} \text{ (the units are J K}^{-1}\text{)}$$

3 An enthalpy change, ΔH, is the heat released or taken in during any change in a system, provided that the system is kept at constant pressure.

4 The following table shows a comparison between exothermic and endothermic reactions.

Reaction	Exothermic	Endothermic
Heat exchange with the surroundings	Heat is released to the surroundings.	Heat is taken in from the surroundings.
Equation of enthalpy change (ΔH)	$\Delta H = H_p - H_r$	
Sign convention for ΔH	$H_p < H_r$ $\therefore \Delta H < 0$ (negative)	$H_p > H_r$ $\therefore \Delta H > 0$ (positive)
Enthalpy level diagram		

Have you mastered?

5 The standard enthalpy change of reaction, $\Delta H_r^{\ominus}$, is the enthalpy change when molar quantities of reactants as stated in the specified equation react together under standard conditions.

6 The standard enthalpy change of formation of a substance, $\Delta H_f^{\ominus}$, is the enthalpy change when one mole of the substance is formed from its elements in their standard states.

7 The standard enthalpy change of combustion of a substance, $\Delta H_c^{\ominus}$, is the enthalpy change when one mole of the substance is completely burnt in oxygen under standard conditions.

8 The standard enthalpy change of neutralization, $\Delta H_n^{\ominus}$, is the enthalpy change when an acid reacts with an alkali to form one mole of water under standard conditions.

9 Follow the steps listed below to calculate the enthalpy change of neutralization from experimental results.

1 Calculate the amount of heat released during the reaction → 2 Calculate the number of moles of water formed → 3 Calculate the enthalpy change of neutralization in kJ mol^{-1}

10 Follow the steps listed below to calculate the enthalpy change of combustion from experimental results.

1 Calculate the amount of heat released during the combustion → 2 Calculate the number of moles of fuel burnt → 3 Calculate the enthalpy change of combustion in kJ mol^{-1}

Have you mastered?

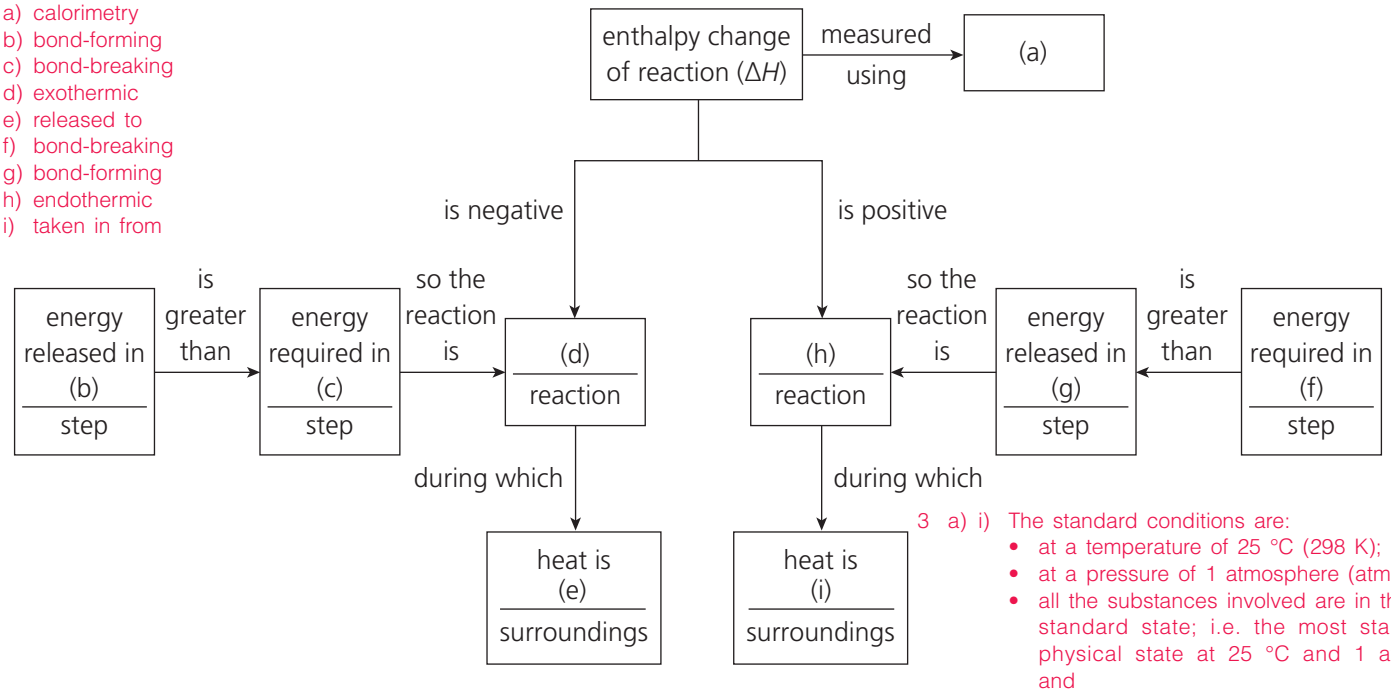
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

a) calorimetry
b) bond-forming
c) bond-breaking
d) exothermic
e) released to
f) bond-breaking
g) bond-forming
h) endothermic
i) taken in from



2 Write the equation, including state symbols, which corresponds to each of the following enthalpy changes.

a) Standard enthalpy change of formation of butane, C_4H_{10}
$4C(graphite) + 5H_2(g) \longrightarrow C_4H_{10}(g)$

b) Standard enthalpy change of combustion of butane, C_4H_{10}
$C_4H_{10}(g) + \frac{13}{2}O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(l)$

c) Standard enthalpy change of formation of glucose, $C_6H_{12}O_6$
$6C(graphite) + 6H_2(g) + 3O_2(g) \longrightarrow C_6H_{12}O_6(s)$

d) Standard enthalpy change of combustion of glucose, $C_6H_{12}O_6$
$C_6H_{12}O_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$

3 a) State the meaning of the terms below:

i) standard conditions;

ii) standard enthalpy change of formation of a compound.

b) i) Write the equation which corresponds to the enthalpy change of formation of liquid water.

ii) Explain why the $\Delta H_f^{\ominus}$ for liquid water is identical to the standard enthalpy change of combustion of hydrogen.

3 a) i) The standard conditions are:
- at a temperature of 25 °C (298 K);
- at a pressure of 1 atmosphere (atm);
- all the substances involved are in their standard state; i.e. the most stable physical state at 25 °C and 1 atm; and
- all the solutions involved have a concentration of 1 mol dm^{-3}.

3 a) ii) The standard enthalpy change of formation of a compound is the enthalpy change when one mole of the compound is formed from its elements in their standard states.

b) i) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$

ii) The two different terms refer to the same thermochemical equation with the same ΔH value.

Part II Multiple choice questions

4 The same amount of heat is added to 50 g samples of each of the metals below. Which metal will have the GREATEST temperature change?

	Metal	Specific heat capacity ($J\ g^{-1}\ K^{-1}$)
A	Aluminium	0.90
B	Copper	0.39
C	Iron	0.45
(D)	Gold	0.13

Gold has the lowest specific heat capacity. With the same amount of heat added, it will have the greatest temperature change.

5 The specific heat capacity of aluminium is $0.90\ J\ g^{-1}\ K^{-1}$.

What is the amount of heat required to raise the temperature of 40.0 g of aluminium from 20.0 °C to 32.3 °C?

A 110 J
B 276 J
(C) 443 J
D 720 J

Amount of heat required
= 40.0 g x 0.90 J g^{-1} K^{-1} x (32.3 – 20.0) K
= 443 J

6 The principle of a firework-type explosion can be demonstrated by igniting a sucrose jelly-baby with sodium chlorate ($NaClO_3$).

The thermochemical equation for the explosion reaction is:

$$2NaClO_3(s) + C_{12}H_{22}O_{11}(s) \longrightarrow 2NaCl(s) + 9C(s) + 3CO_2(g) + 11H_2O(g) \qquad \Delta H = -2\ 192\ kJ$$

What is the amount of heat released when one jelly-baby containing 5.13 g of sucrose ($C_{12}H_{22}O_{11}$) is exploded?

(Molar mass of $C_{12}H_{22}O_{11}$ = 342.0 g mol^{-1})

A 12.5 kJ
(B) 32.9 kJ
C 112 kJ
D 146 kJ

According to the thermochemical equation, the explosion reaction of 1 mole of sucrose releases 2 192 kJ of heat.
Number of moles of sucrose exploded
$= \frac{5.13\ g}{342.0\ g\ mol^{-1}}$
= 0.0150 mol
Heat released from 5.13 g of sucrose
= 0.0150 mol x 2 192 kJ mol^{-1}
= 32.9 kJ
∴ 32.9 kJ of heat are released when one jelly-baby is exploded.

7 Which of the following pairs of reactants would react in water to give out the largest amount of heat?

A 1 mol of HCl and 1 mol of KOH
(B) 1 mol of H_2SO_4 and 2 mol of KOH
C 1 mol of $(COOH)_2$ and 2 mol of KOH
D 1 mol of CH_3COOH and 1 mol of KOH

(HKDSE, Paper 1A, 2012, 14)

8 Consider the following thermochemical equation:

$$C_2H_5OH(g) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H = -1\ 368\ kJ\ mol^{-1}$$

The ΔH value for
$4CO_2(g) + 6H_2O(l) \longrightarrow 2C_2H_5OH(g) + 6O_2(g)$
would be

A +684 kJ.
B +1 368 kJ.
(C) +2 736 kJ.
B +5 472 kJ.

9 The meals-ready-to-eat in the military can be heated on a flameless heater. The energy in the heater comes from the following reaction:

$$Mg(s) + 2H_2O(l) \longrightarrow Mg(OH)_2(s) + H_2(g) \qquad \Delta H^{\ominus} = -353\ kJ$$

What is the mass of magnesium needed to supply the heat to warm 25.0 g of water from 25.0 °C to 85.0 °C?

(Relative atomic mass: Mg = 24.3; specific heat capacity of water = 4.18 J g^{-1} K^{-1})
Refer to page T17 for explanations.

A 0.0178 g
(B) 0.433 g
C 17.8 g
D 43.3 g

8 When we reverse an equation, the magnitude of ΔH for the equation remains the same, but its sign changes.
∴ $2CO_2(g) + 3H_2O(l) \longrightarrow C_2H_5OH(g) + 3O_2(g)$ $\Delta H = +1\ 368\ kJ\ mol^{-1}$
When we multiply both sides of an equation by a factor, ΔH must also be multiplied by the same factor.
∴ $4CO_2(g) + 6H_2O(l) \longrightarrow 2C_2H_5OH(g) + 6O_2(g)$ $\Delta H = 2(+1\ 368\ kJ) = +2\ 736\ kJ$

10 Which equation represents the reaction for which the enthalpy change is the standard enthalpy change of formation, $\Delta H_f^{\ominus}$, of sodium nitrate, $NaNO_3$?

A $2Na(s) + N_2(g) + 3O_2(g) \longrightarrow 2NaNO_3(s)$

(B) $Na(s) + \frac{1}{2}N_2(g) + \frac{3}{2}O_2(g) \longrightarrow NaNO_3(s)$

C $Na(s) + N(g) + 3O(g) \longrightarrow NaNO_3(s)$

D $Na(g) + \frac{1}{2}N_2(g) + \frac{3}{2}O_2(g) \longrightarrow NaNO_3(g)$

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2010, 4)

11 The molar enthalpy change of combustion of some alkanes is given below in kJ mol^{-1}.

C_3H_8	–2 219
C_4H_{10}	–2 877
C_5H_{12}	–3 509
C_6H_{14}	–4 163

Another alkane was found to have an enthalpy change of combustion of –6 125 kJ mol^{-1}. The alkane is

A C_7H_{16}.

B C_8H_{18}.

(C) C_9H_{20}.

D $C_{10}H_{22}$.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2012, 8)

12 The following table lists some information on mixing hydrochloric acid with alkalis:

Mixture	Temperature rise (°C)
25 cm^3 of 1 mol dm^{-3} HCl(aq) + 25 cm^3 of 1 mol dm^{-3} NH_3(aq)	*x*
25 cm^3 of 1 mol dm^{-3} HCl(aq) + 25 cm^3 of 1 mol dm^{-3} NaOH(aq)	*y*

Refer to page T17 for explanations.

Which of the following concerning the values of temperature rise is correct?

A $x = y$

B $x > y$

(C) $x < y$

D $x = 2y$

13 In an experiment to determine the enthalpy change of combustion of an alcohol, a spirit burner containing the alcohol was weighed, lit and placed under a copper can containing a known volume of water. The temperature rise of the water was measured and the burner re-weighed. The enthalpy change calculated from the results was much less exothermic than the value reported in the literature.

Which of the following factors is most likely to be the cause of this error?

(A) Heat loss around the side of the copper can

B The use of a thermometer with a range of 0–110 °C rather than 0–50 °C

C The use of a measuring cylinder for measuring the water rather than a pipette

D Evaporation of the alcohol during the weighing

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 4)

14 Which of the following process(es) is / are exothermic?

(1) Dissolving ammonium nitrate in water
(2) Respiration
(3) Sublimation of dry ice

A (1) only

(B) (2) only

C (1) and (3) only

D (2) and (3) only

(1) Dissolution of ammonium nitrate is an endothermic process and is used in cold packs. When using the cold pack, ammonium nitrate dissolves in water and the temperature of the water drops.

(3) Heat is absorbed in the sublimation of dry ice. It is an endothermic process.

Unit Exercise

10 In the equation of Option A, 2 moles of sodium nitrate are formed.
In the left side of the equation of Option C, the standard states of nitrogen and oxygen should be $N_2(g)$ and $O_2(g)$.
In the left side of the equation of Option D, the standard state of sodium should be Na(s).

11

Alkane	Enthalpy change of combustion (kJ mol^{-1})	Difference in enthalpy change of combustion (kJ mol^{-1})
C_3H_8	–2 219	658
C_4H_{10}	–2 877	632
C_5H_{12}	–3 509	654
C_6H_{14}	–4 163	

From the above data, it can be deduced that the difference of enthalpy change of combustion of two neighbouring alkanes is about 600 kJ mol^{-1}.
The difference of the enthalpy changes of combustion of another alkane and C_6H_{14} is
$[-4\ 163 - (-6\ 125)]$ kJ mol^{-1} = 1 962 kJ mol^{-1}
$\frac{1\ 962}{600} \approx 3$
The difference of 1 962 kJ mol^{-1} is equivalent to 3 $-CH_2-$ groups, therefore the alkane should be C_9H_{20}.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

15 947 J are required to raise the temperature of 125 g of stainless steel by 15.7 °C. What is the specific heat capacity of the stainless steel?

16 Thermit reactions broadly refer to exothermic oxidation-reduction reactions between a metal powder and a metal oxide. One example is the reaction of finely divided iron(III) oxide with aluminium powder. This reaction results in a very high temperature, and is commonly used in the welding of rail tracks for trains. At this very high temperature, the molten iron formed joins the rail tracks together.

a) Complete and balance the chemical equation for the following thermit reaction.

$$____Fe_2O_3(s) + ____Al(s) \longrightarrow ____ + ____$$

b) Sketch a labelled enthalpy level diagram for this reaction.

(HKDSE, Paper 1B, 2013, 7(a))

17 When solutions of potassium carbonate and calcium chloride are mixed together, the following reaction takes place:

$$CaCl_2(aq) + K_2CO_3(aq) \longrightarrow CaCO_3(s) + 2KCl(aq)$$

An experiment was carried out to measure the enthalpy change of this reaction. 50.0 cm^3 of 2.00 mol dm^{-3} potassium carbonate solution were added to 50.0 cm^3 of 2.00 mol dm^{-3} calcium chloride solution. The temperature fell by 3.2 °C.

Refer to page T17.

a) Draw a labelled diagram of the experimental set-up that could have been used to carry out the experiment.

b) Calculate the enthalpy change of the reaction, in kJ mol^{-1}.

+13.4 kJ mol^{-1}

(Density of resulting solution = 1.00 g cm^{-3}; specific heat capacity of resulting solution = 4.18 J g^{-1} K^{-1})

c) If the experiment was repeated using 100.0 cm^3 of each solution, predict what the fall in temperature would be.

The fall in temperature remains unchanged (i.e. it is still 3.2 °C).

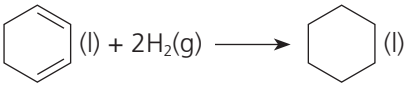
15 Amount of heat required = $m \times c \times \Delta T$

$947 \text{ J} = 125 \text{ g} \times c \times 15.7 \text{ K}$

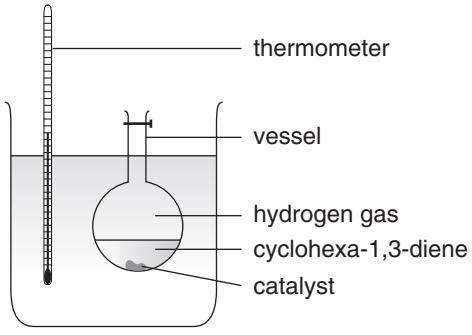
$c = 0.483 \text{ J g}^{-1} \text{ K}^{-1}$

∴ the specific heat capacity of stainless steel is 0.483 J g^{-1} K^{-1}.

18 An experiment was carried out to determine the enthalpy change of hydrogenation of cyclohexa-1,3-diene (C_6H_8).



12.0 g of cyclohexa-1,3-diene were treated with excess hydrogen gas in the presence of a catalyst in the vessel shown below. The vessel was immersed in 500.0 g of water.



At the end of the experiment, the temperature of the water increased by 15.4 °C.

Refer to page T18.

a) Calculate the enthalpy change of hydrogenation of cyclohexa-1,3-diene, in kJ mol^{-1}, under the experimental conditions.

–215 kJ mol^{-1}

b) Suggest TWO assumptions in your calculation.

(Relative atomic masses: H = 1.0, C = 12.0; specific heat capacity of water = 4.18 J g^{-1} K^{-1})

19 Bottled propane is used as a fuel for heating. The equation for its complete combustion is:

$$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l) \qquad \Delta H = -2\,220 \text{ kJ mol}^{-1}$$

Calculate the heat evolved when 176.0 g of propane are completely burnt in air.

(Relative atomic masses: H = 1.0, C = 12.0)

(Edexcel GCE O Level, Paper 2, Jan. 2011, 4(a)(iii))

19 Number of moles of propane $= \dfrac{176.0 \text{ g}}{(3 \times 12.0 + 8 \times 1.0) \text{ g mol}^{-1}}$

$= 4.00 \text{ mol}$

Heat evolved when 176.0 g of propane are burnt in air

= 2 220 kJ mol^{-1} x 4.00 mol

= 8 880 kJ

20 The table below lists the standard enthalpy changes of neutralization of three acids with sodium hydroxide solution.

Acid	$\Delta H_n^{\ominus}$ (kJ mol^{-1})
HCl(aq)	–57.1
HNO_3(aq)	–57.1
CH_3COOH(aq)	–55.2

Refer to next page.

a) Define the term 'standard enthalpy change of neutralization'.

b) Account for the following statements:

i) The standard enthalpy change of neutralization of HCl(aq) with NaOH(aq) is the same as that of HNO_3(aq) with NaOH(aq).

ii) The standard enthalpy change of neutralization of HCl(aq) with NaOH(aq) is more exothermic than that of CH_3COOH(aq) with NaOH(aq).

21 Enthalpy changes of combustion, ΔH_c, are amongst the easiest enthalpy changes to determine directly.
Refer to next page.

a) Define the term 'enthalpy change of combustion'.

b) A student carried out an experiment to determine the enthalpy change of combustion of pentan-1-ol, $CH_3(CH_2)_4OH$.

In the experiment, 1.76 g of pentan-1-ol were burnt. The heat was used to heat 250.0 cm^3 of water from 24.0 °C to 78.0 °C.

i) Calculate the heat released, in kJ, during the combustion of 1.76 g of pentan-1-ol.

(The specific heat capacity of water = 4.18 J g^{-1} K^{-1}, density of water = 1.00 g cm^{-3})

ii) Calculate the amount, in moles, of pentan-1-ol that was burnt.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

iii) Calculate the enthalpy change of combustion of pentan-1-ol.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2009, 2(a)–(b))

22 A group of students devised an experiment which they believed would enable them to investigate the strength of the intermolecular forces between ethyl ethanoate molecules ($CH_3COOCH_2CH_3$) and trichloromethane molecules ($CHCl_3$).

They mixed exactly 0.100 mole of each of the two liquids in a copper calorimeter and recorded the following results. The starting temperature of both liquids was the same.

Mass of 0.100 mol of ethyl ethanoate (g)	8.80
Mass of 0.100 mol of trichloromethane (g)	11.95
Increase in temperature (ΔT) on mixing (K)	9.5

Refer to page T19.

a) i) Write an expression for the heat change (q) which relates mass (m), specific heat capacity (c) and change in temperature (ΔT).
$q = m \times c \times \Delta T$

ii) Calculate the amount of heat required to increase the temperature of 8.80 g of ethyl ethanoate by 9.5 K during the mixing process. (You should assume that c for ethyl ethanoate = 1.92 J g^{-1} K^{-1})
161 J

iii) Calculate the amount of heat required to increase the temperature of 11.95 g of trichloromethane by 9.5 K during the mixing process. (You should assume that c for trichloromethane = 0.96 J g^{-1} K^{-1})
109 J

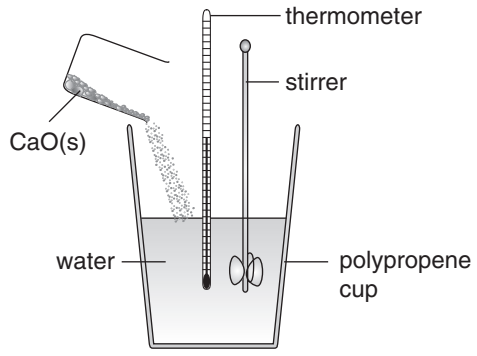
iv) Using the values from parts (a)(ii) and (iii), calculate the molar enthalpy change in kJ mol^{-1} for the mixing process.
–2.70 kJ mol^{-1}

b) The students deduced that the heat change was due only to the formation of intermolecular forces between ethyl ethanoate molecules and trichloromethane molecules.

Ignoring all experimental errors, give ONE reason why the students may have made an incorrect deduction.

(AQA Advanced Subsidiary GCE, Unit 2, Jun. 2009, 3)

23 a) A student carried out an experiment to determine the enthalpy change of the reaction of calcium oxide with water. The set-up used is shown in the diagram below:



The experimental results are as follows:

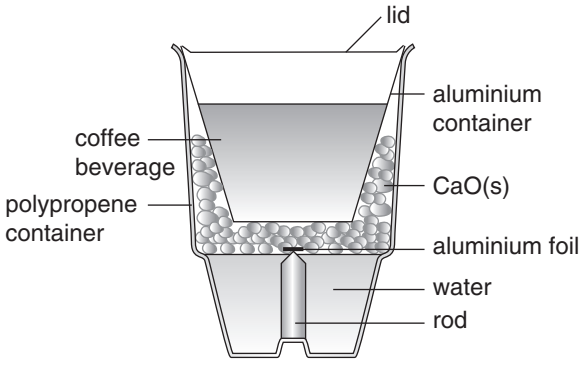
Mass of CaO(s) used	3.0 g
Volume of water in the cup	50.0 cm^3
Initial temperature of water in the cup	28.2 °C
Highest temperature attained by the $Ca(OH)_2(aq)$ formed	46.7 °C

i) Calculate the enthalpy change, in kJ mol^{-1}, of the reaction of calcium oxide with water under the conditions of the experiment.

(Assume: density of water is 1.0 g cm^{-3} and specific heat capacity of the $Ca(OH)_2(aq)$ formed is 4.2 J g^{-1} K^{-1}; the polypropene cup, thermometer and stirrer used all have negligible heat capacity.)

ii) According to the literature, $\Delta H^{\ominus}$ for this reaction is –82.2 kJ mol^{-1}. Suggest ONE reasonable explanation for the discrepancy between the literature value and the value obtained in (i).

b) The diagram below shows the design of a can of self-heating coffee beverage. When the bottom of the can is pushed, the rod will pierce the aluminium foil and cause mixing of the water and calcium oxide. The coffee beverage in the can will then be heated up.



i) With reference to the properties of the materials involved, explain why

(I) a polypropene container is used to contain the calcium oxide, and

(II) an aluminium container is used to contain the coffee beverage.

ii) Suggest ONE reasonable explanation for using calcium oxide in this type of self-heating beverage can.

(HKDSE, Practice paper 1B, 2012, 7)

20 a) The standard enthalpy change of neutralization, $\Delta H_n^{\ominus}$, is the enthalpy change when an acid reacts with an alkali to form one mole of water under standard conditions.

b) i) Both neutralization reactions involve the reaction between a strong monobasic acid and NaOH(aq).
The same chemical change occurs in both cases.
$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$

ii) Ethanoic acid is a weak acid.
In the course of neutralization, molecules of ethanoic acid consume energy for dissociation.

21 a) Standard enthalpy change of combustion of a substance is the enthalpy change when one mole of the substance is completely burnt in oxygen under standard conditions.

b) i) Heat released during the combustion of pentan-1-ol
= 250.0 g x 4.18 J g^{-1} K^{-1} x (78.0 – 24.0) K
= 56 400 J
= 56.4 kJ

ii) Number of moles of pentan-1-ol burnt
$= \dfrac{1.76 \text{ g}}{(5 \times 12.0 + 12 \times 1.0 + 16.0) \text{ g mol}^{-1}}$
= 0.0200 mol

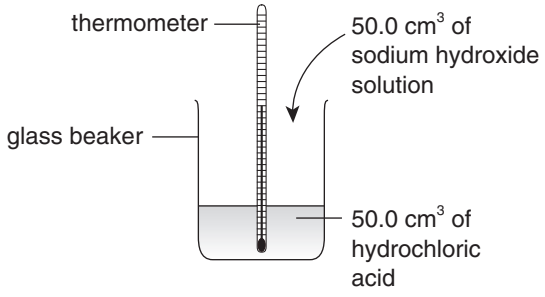
iii) $\Delta H_c[C_5H_{11}OH(l)] = \dfrac{-56.4 \text{ kJ}}{0.0200 \text{ mol}}$
$= -2\,820 \text{ kJ mol}^{-1}$

24 Read the information about enthalpy changes and then answer the questions.

A student did an experiment to find the enthalpy change when hydrochloric acid reacts with sodium hydroxide solution. The equation which represents the reaction is:

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

The student used the apparatus shown in the diagram.



The student placed 50.0 cm^3 of hydrochloric acid in a glass beaker and measured the temperature. The student then quickly added 50.0 cm^3 of sodium hydroxide solution and stirred the mixture with the thermometer. The highest temperature was recorded. The student repeated the experiment, and calculated the temperature change each time.

	Experiment 1	Experiment 2	Experiment 3	Experiment 4
Initial temperature (°C)	19.0	22.0	19.2	19.0
Highest temperature (°C)	26.2	29.0	26.0	23.5
Temperature change (°C)	7.2	7.0	6.8	4.5

Unit Exercise

a) The biggest error in this experiment is heat loss. Suggest how the apparatus could be modified to reduce heat loss.

a) Any one of the following:
- Use an expanded polystyrene cup.
- Insulate the beaker.
- Add a lid.

b) Suggest why it is important to stir the chemicals thoroughly.
To ensure that the chemicals react completely.

c) Which ONE of these experiments was probably carried out on a different day to the others? Explain your answer. Experiment 2.
The initial temperature is different.

d) Suggest why *Experiment 4* should NOT be used to calculate the average temperature change.
The temperature change of this experiment is an outlier.

e) Calculate the average temperature change from the first three experiments. $\frac{(7.2 + 7.0 + 6.8)\ °C}{3} = 7.0\ °C$

f) Use the following equation to calculate the enthalpy change for this reaction.

enthalpy change (joules) = 100.0 x 4.2 x average temperature change

Enthalpy change = (100.0 x 4.2 x 7.0) J
= 2 940 J

g) Which one of these enthalpy level diagrams, A or B, represents the enthalpy change for this reaction? Explain why.

diagram A

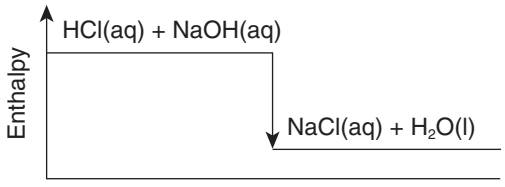


diagram B

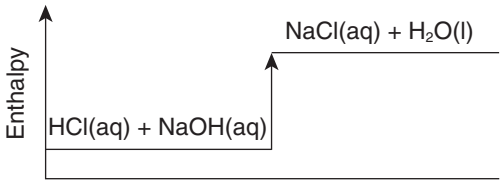


Diagram A
The reaction between HCl(aq) and NaOH(aq) is exothermic.
The total enthalpy of the products is less than that of the reactants.

(AQA GCSE (Higher Tier), Chemistry, Unit 3, Jan. 2009, 2)

25 The enthalpy change for the reaction between zinc and copper(II) sulphate solution was determined using the procedure below. The ionic equation for the reaction is

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

Procedure

1 Weigh about 5 g of zinc powder.

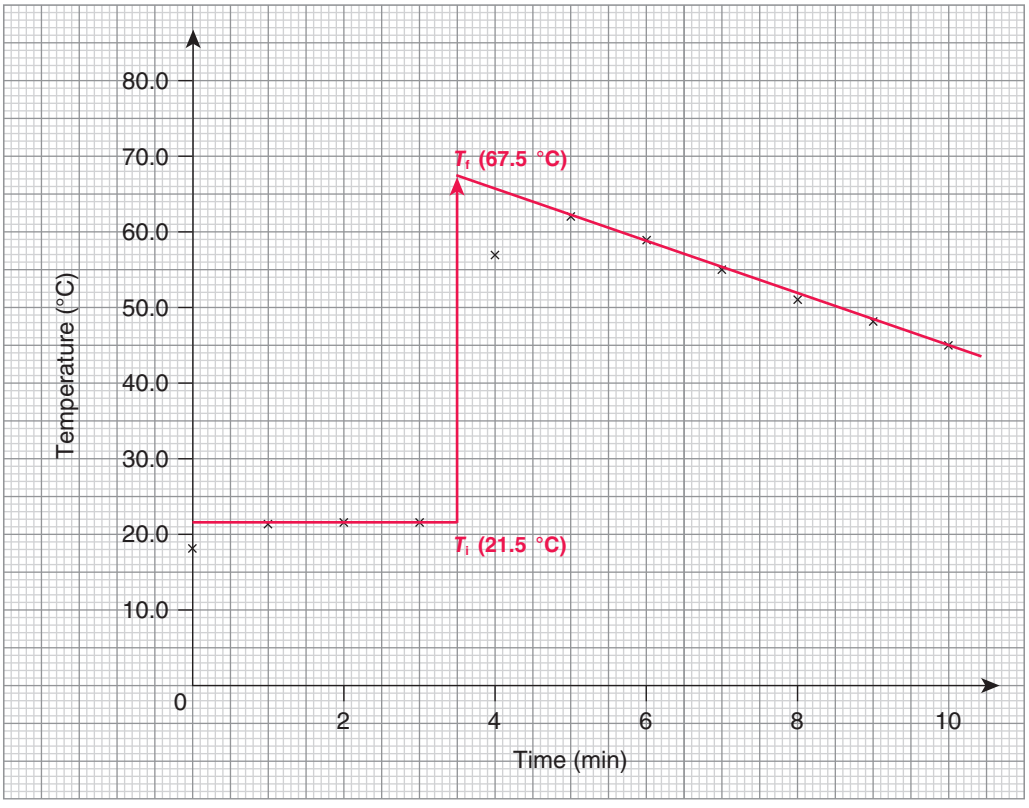
2 Measure 50.0 cm^3 of 1.00 mol dm^{-3} copper(II) sulphate solution into a polystyrene cup.

3 Stir the solution continuously with a thermometer and measure the temperature of the solution each minute for 3 minutes.

4 At exactly 3.5 minutes add the zinc powder to the copper(II) sulphate solution.

5 Continue to stir the solution and read the temperature each minute from 4 to 10 minutes.

The results obtained are shown in the graph below.



a) i) Use the graph to estimate the maximum temperature change, ΔT, for the reaction. Show your working on the graph.

ii) Use the value you have obtained for ΔT to calculate the heat produced in the reaction between zinc and copper(II) sulphate.

(Assume the specific heat capacity of the solution to be 4.18 J g^{-1} K^{-1} and the density of the solution to be 1.00 g cm^{-3}.)

iii) Calculate the enthalpy change, ΔH, for this reaction. Your answer should be in units of kJ mol^{-1}.

b) i) Explain why the temperature of the solution is measured for 3 minutes before adding the zinc.

ii) Explain why the temperature of the solution is measured over a period of time after adding the zinc.

iii) A polystyrene cup is used, rather than a glass beaker, to reduce heat loss. Explain why a polystyrene cup is a good choice.

iv) Explain why the solutions are continuously stirred in this experiment.

v) Suggest a piece of apparatus suitable for measuring the 50.0 cm^3 of copper(II) sulphate solution in this experiment.

vi) Suggest ONE change in the apparatus used (other than using more accurate measuring equipment) that would improve the accuracy of the results.

c) In two further experiments, using more accurate equipment and an improved method, the ΔH for the reaction between zinc and copper(II) sulphate was determined to be –216.8 kJ mol^{-1} while that for the reaction between zinc and lead(II) nitrate was found to be –154.0 kJ mol^{-1}.

Place the three metals (copper, lead and zinc) in order of decreasing reactivity and justify your answer by using these ΔH values.

(Edexcel Advanced Subsidiary GCE, Unit 3B, Jun. 2009, 2(a)–(c))

Unit Exercise

25 a) i) Maximum temperature change for the reaction = (67.5 – 21.5) °C
= 46.0 °C

ii) Heat produced in the reaction = 50.0 g x 4.18 J g^{-1} K^{-1} x 46.0 K
= 9 610 J

iii) Number of moles of $CuSO_4$ undergoing the reaction = 1.00 mol dm^{-3} x $\frac{50.0}{1\ 000}$ dm^3
= 0.0500 mol

$\Delta H = \frac{-9\ 610\ \text{J}}{0.0500\ \text{mol}}$
= –192 kJ mol^{-1}

b) i) To ensure a steady temperature.
ii) To perform cooling correction / to compensate the heat loss.
iii) Low heat capacity / good insulator of heat
iv) To ensure uniform temperature in the reaction mixture.
v) Burette / pipette
vi) Cover the polystyrene cup with a lid. / Improve insulation.

c) Zinc > lead > copper
The more exothermic / more negative the ΔH, the more reactive the metal.